Of the Standards of Science

Computational Modelling as Psychology's Common Language

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2026-05-08

Abstract

The replication crisis in psychology has been attributed to incentive structures and statistical norms; the role of researcher skill distribution has not been formally modelled. This thesis presents an agent-based model of a stylised scientific community of fifty researchers across ten laboratories and ten research domains, producing and evaluating scientific models over 300 simulated time steps under evolutionary selection. Two scenarios are compared under otherwise identical conditions: a *Specialist* community, in which researchers have one pronounced domain expertise and weak secondary skills, and a *Generalist* community, in which they maintain a meaningful baseline across all domains alongside their primary expertise. Across 30 independent runs per scenario, the specialist community produced a higher replication failure rate (.62 vs. .59; $d = 0.78$, $p = .006$) and substantially greater concentration of publications across domains (Gini .139 vs. .093; $d = 2.14$, $p < .0001$); two exploratory hypotheses about reputation and debunking vulnerability did not reach significance. Three methodological analyses sharpen the interpretation: a Uniform-skill control reframes H1 as U-shaped rather than monotonic in skill specialisation; leave-one-out parameter recovery confirms the manipulation is identifiable from outcomes; and a layer-decomposition ablation shows the H1 effect requires interaction between skill heterogeneity and career-pressure dynamics, refuting the strongest architectural-tautology critique. Directional results follow from how skill enters action selection and replication probability; they are best read as quantifications of mechanism magnitude under a particular formalisation, not as evidence about real scientific communities. The model's contribution is to establish that the *communicative infrastructure* of a discipline — how broadly researchers can evaluate work outside their primary expertise — is a tractable lever for formal modelling alongside incentive structures.

Keywords: replication crisis, agent-based modelling, skill specialisation, publication bias, scientific communities

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Introduction

This thesis investigates a structural explanation for the replication crisis in psychology: the degree of skill specialisation among researchers. While prior work has examined how statistical norms (Smaldino & McElreath, 2016), incentive structures (Nissen et al., 2016), and methodological diversity (Devezer et al., 2019; Weisberg & Muldoon, 2009) shape scientific communities, the role of domain-level skill distribution — how narrowly or broadly researchers are trained across research areas — has not been formally modelled. To address this gap, an agent-based computational model is developed in which a stylised scientific community of fifty researchers, organised across ten laboratories and ten research domains, produces and evaluates scientific models over time. Two contrasting scenarios are compared: a *Specialist* community, in which researchers have a single pronounced peak skill and weak secondary competencies, and a *Generalist* community, in which researchers maintain a meaningful baseline across multiple domains alongside their primary expertise. All other structural parameters are held constant.

Four outcomes are examined across 30 independent simulation runs per scenario. Two outcomes — replication failure rate and domain concentration of the publication landscape — were pre-specified from theory before simulations were run and constitute the confirmatory core of the analysis. Two further outcomes — researcher reputation accumulation and vulnerability to debunking — were identified during model development and are treated as exploratory. The model does not aim to reproduce any specific empirical dataset; rather, it tests whether the structural contrast between specialist and generalist skill distributions is sufficient to generate differences in systemic outcomes consistent with known features of the replication crisis (Ioannidis, 2005; Open Science Collaboration, 2015). Results are assessed using non-parametric tests with effect size estimates, and the robustness of findings is examined through a one-at-a-time sensitivity analysis.

The Replication Crisis and Its Roots

Psychology is in a period of acute self-examination. The most visible symptom of this is the replication crisis: a large-scale coordinated effort by the Open Science Collaboration found that fewer than half of a representative sample of published psychological findings replicated when tested independently (Open Science Collaboration, 2015), and subsequent multi-site replication projects have produced converging estimates (Camerer et al., 2018; Klein et al., 2014). This is not a problem unique to psychology — Ioannidis (2005) demonstrated formally that when base rates of true hypotheses are low and statistical power is modest, the majority of nominally significant published findings are expected to be false — but it has been felt with particular force in a discipline that derives much of its applied authority from the reliability of its empirical claims.

Two classes of explanation dominate the discussion. The first concerns incentive structures: academic careers are advanced by publications, and publications are advanced by significant results, which creates systematic pressure toward the kind of flexible analysis, selective reporting, and underpowered studies that inflate false-positive rates (Holman & Bruner, 2017; Nissen et al., 2016; Smaldino & McElreath, 2016). Anonymous-survey evidence indicates that questionable research practices are widespread among practising psychologists (John et al., 2012), consistent with a community-level dynamic in which individually rational career behaviour produces collectively unreliable findings. The second, and arguably more fundamental, concerns the structure of psychological theorising itself. As Meehl (1978) argued decades before the replication crisis became a household term, the slow progress of soft psychology is rooted in the use of verbal theories that are flexible enough to accommodate almost any pattern of data after the fact. When a theory can be rendered consistent with contradictory findings by adjusting its informal interpretation, it generates no genuine predictions and cannot be falsified. It accumulates asterisks rather than knowledge.

These two problems are interrelated in a way that is easy to overlook. Verbal theories are not merely imprecise — they are structurally invisible to researchers in other labs. When the mechanism linking two variables is described only in natural language, two researchers who believe they are studying the same phenomenon may operationalise it so differently that replication is structurally impossible even before any incentive problem arises. A shared formal language would not eliminate researcher degrees of freedom, but it would make theoretical commitments explicit enough to be auditable, comparable, and cumulatively buildable. As Smaldino (2017) argues, computational models force the scientist to commit: assumptions are specified, parameters are quantified, and predictions follow mechanically rather than rhetorically. The very rigidity that makes a computational model seem limiting is what gives it communicative power.

The case for formal models, however, goes beyond the individual researcher. McElreath & Smaldino (2015) demonstrate through a population-level mathematical model that the same statistical parameters — low base rate, low power, high false-positive rate — that make individual findings unreliable also make individual replications unreliable. Under these conditions, not one but multiple successful replications are required before a result carries strong evidential weight, because replication can only function as an epistemological filter if the underlying detection process is reliable. Crucially, the dynamics of scientific communication — which findings get published, which replications are attempted, and how results diffuse through a field — determine whether the community converges on truth regardless of individual good intentions.

A related line of agent-based work, beginning with Zollman (2007)'s analysis of communication structure in epistemic communities, shows that the topology of who-talks-to-whom shapes whether a research community converges on the correct theory; sparser networks can outperform denser ones when individual signals are noisy, because they allow minority views the time to accumulate corroborating evidence. Devezer et al. (2019) extend this line in a directly relevant direction. Using an agent-based model of a scientific community, they show that reproducibility and truth convergence are not equivalent properties, and that the *structure* of the research population matters as much as individual researcher behaviour. Populations dominated by conservative researchers who test only incremental variants of the current consensus achieve high reproducibility while spending little time at the true model. Populations with greater methodological diversity — including researchers willing to explore distant regions of model space — discover the true model faster and are more robust to getting stuck in local optima. Importantly, no single research strategy is uniformly optimal: epistemic diversity, rather than any particular method, is what buffers a scientific community against its own structural limitations (Weisberg & Muldoon, 2009).

The Theory Crisis: Why Formalization Alone Is Insufficient

The discussion so far might suggest that the remedy for psychology's problems is straightforward: replace verbal theories with formal ones. Eronen & Bringmann (2021) — alongside Oberauer & Lewandowsky (2019) — complicate this picture in an important way. They argue that psychology faces a theory crisis distinct from and deeper than the replication crisis: even in a hypothetical world where every finding replicated perfectly, psychological theories would still fail to cumulate, because the phenomena they describe are not stable enough to anchor formal models.

Three structural problems compound each other. First, psychological interventions are *fat-handed*: a manipulation designed to induce, say, cognitive load simultaneously affects arousal, motivation, working memory, and mood, making it impossible to isolate the mechanism of interest. Second, construct validity is chronically weak — the same label (e.g., "working memory", "self-esteem") refers to operationally distinct things across labs, so replication is structurally ambiguous even when it succeeds. Third, and most fundamentally, psychology lacks the bedrock of robust, theory-independent phenomena from which formal theories could be bottom-up constructed. Natural sciences progress in part because anomalies are sharp — a two-degree discrepancy in a measured constant is a crisis. In psychology, the effect sizes themselves are contested.

This is a genuine challenge for any argument in favour of computational modelling, including the one made in this thesis. The position taken here is nevertheless that Eronen & Bringmann (2021)'s critique sharpens rather than defeats the case for formal models. A computational model cannot conjure stable phenomena from an unstable empirical record, but it can make the *assumptions* about stability explicit — and explicit assumptions can be tested, refined, and challenged across labs in a way that implicit verbal assumptions cannot. This thesis does not argue that psychology can or should become a natural science in the classical sense; it argues that, whatever the ultimate stability of its phenomena, the communicative infrastructure of a discipline determines whether it converges on its best available models, and that computational formalisation improves that infrastructure.

Computational Models as a Common Language

Taken together, the findings reviewed above suggest that the replication crisis in psychology is not primarily a story of individual bad actors but of a field whose theoretical and communicative infrastructure is too informal to support cumulative progress. The appropriate response is not only procedural reform — pre-registration, power analysis, open data — but a deeper shift toward shared formal frameworks that make theoretical commitments visible, comparable, and testable across labs (Muthukrishna & Henrich, 2019; Rooij & Baggio, 2021).

The case for such frameworks is partly an argument from the more mature sciences. Physics, mathematics, and computer science share a structural feature that psychology largely lacks: a stock of explicit axioms, formal definitions, and standardised notational conventions that allow distant researchers to interpret one another's claims with minimal ambiguity. A theorem stated in the language of differential geometry travels between laboratories without semantic loss; a verbal construct like "self-control" or "implicit attitude" does not. Muthukrishna & Henrich (2019) argue that the absence of an overarching theoretical framework — comparable to evolutionary theory in biology or general relativity in physics — is itself a structural cause of the replication crisis, because without such a framework empirical findings cannot be evaluated as confirmatory or surprising relative to a shared body of expectations. Rooij & Baggio (2021) develop a complementary point: psychological science has prioritised the *empirical* cycle (testing effects) over the *theoretical* cycle (specifying explanations with the formal precision needed to determine what would count as a test in the first place), and recommend importing the levels-of-analysis methodology that has worked in cognitive science from its informatic roots.

Computational modelling is the most mature candidate for such a framework in psychology and inherits its discipline directly from these older traditions. By demanding that theoretical commitments be expressed in code that runs, it forces the kind of definitional explicitness that verbal theory routinely defers — every assumption must be operationalised, every parameter named, every interaction between mechanisms specified at the level of execution (Guest & Martin, 2021). It provides the quantitative precision that verbal theories lack, the mechanistic explicitness that enables genuine replication, and — as the work reviewed here demonstrates — the capacity to study the dynamics of scientific communities themselves as objects of inquiry. The point is not that psychology should imitate physics in subject matter, but that it can adopt a comparable standard for how theoretical claims are stated, compared, and accumulated.

Research Questions and Model Overview

At its core, the model asks whether restricting researchers' cross-domain competence is, by itself, sufficient to generate higher replication failure and greater publication concentration under otherwise identical institutional conditions. The simulation model consists of three entity types — Labs, Researchers, and Scientific Models — interacting across ten research domains. Figure 1 provides a schematic overview of the model architecture and the causal relationships between its components.

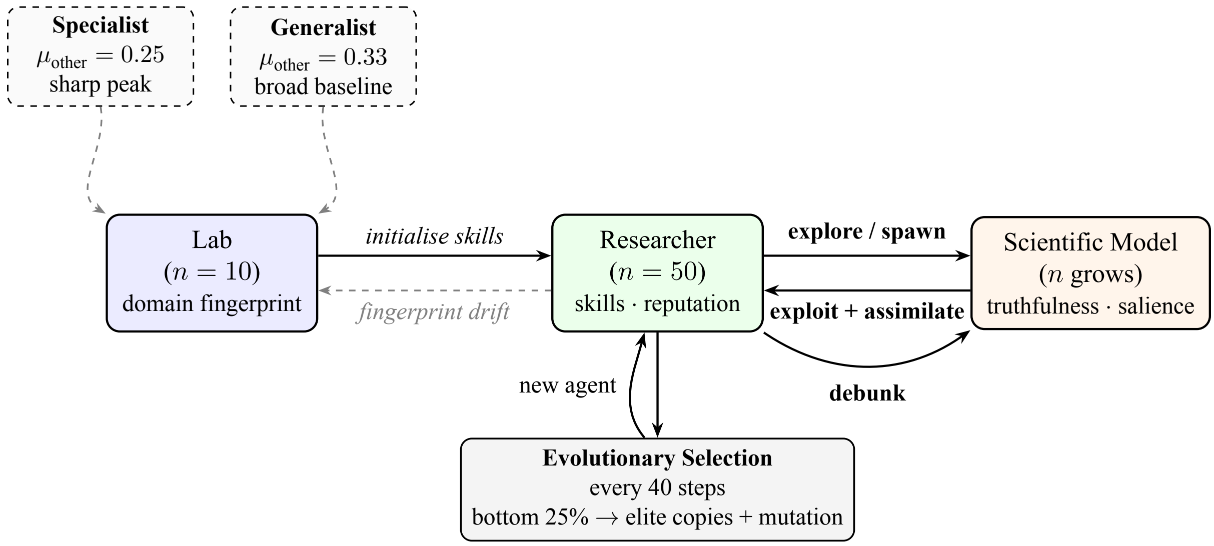

Figure 1
Schematic architecture of the agent-based model. Solid arrows indicate active causal processes; dashed arrows indicate passive or initialisation relationships. The two scenario boxes (top) define the only parameter that differs between conditions: μ_{other}, the mean non-peak skill of lab fingerprints. All other structural parameters are identical across both scenarios.

Four research questions motivated the study design. Two were pre-specified from theory before simulations were run (H1, H2) and constitute the confirmatory core of the analysis. Two emerged during model development and are treated as exploratory (H3, H4).

H1 (Replication failure rate). Does a specialist research community produce a higher rate of replication failure than a generalist community? Specialist researchers have a narrow skill profile, meaning their ability to successfully engage with models outside their primary domain is low. They are therefore more likely to fail replication attempts of models in areas where they lack competence, and less able to identify or correct errors in other domains. It is hypothesised that the specialist scenario will show a systematically higher replication failure rate than the generalist scenario.

H2 (Domain concentration). Does a specialist research community produce greater concentration of publications in fewer research domains? Because specialist researchers preferentially exploit models in their peak domain and have little incentive or ability to work across domains, the publication landscape is expected to become concentrated in a small subset of domains. Generalist researchers, by contrast, can profitably engage with a broader range of domains, producing a more evenly distributed literature. It is hypothesised that the specialist scenario will show a higher Gini coefficient of domain-level publication counts than the generalist scenario.

H3 (Researcher reputation — exploratory). Does community structure affect the average career success of individual researchers, as measured by accumulated reputation? Generalist researchers can accumulate publications across multiple domains and are less exposed to the saturation of any single domain, potentially enabling more consistent reputation growth. This hypothesis is treated as exploratory; the direction is plausible but not strongly constrained by theory.

H4 (Debunking stability — exploratory). Does community structure affect the degree to which researchers lose reputation through debunking events? Specialist researchers concentrate both their expertise and their publications in a single domain; successful debunking of a key model in that domain could therefore inflict proportionally larger reputational losses than the same event would in a generalist community. This hypothesis is treated as exploratory; the direction follows from the logic of concentration but the mechanism involves several countervailing forces that the model will adjudicate.

Methods

The choice of an agent-based model rather than a regression, closed-form equation, or game-theoretic analysis follows from features of the substantive question. The outcomes of interest — replication rates, publication concentration, reputation trajectories — *emerge* from many local interactions among heterogeneous researchers (each with a distinct skill profile and career history) under feedback loops (success breeds attention, attention breeds further success). Closed-form analysis is unavailable for a system with this many interacting state variables; regression treats agents as exchangeable, which the SPEC/GEN manipulation is designed to violate; equilibrium analysis presupposes a steady state that the explicit selection-and-replacement dynamics make ill-defined. ABM is the natural frame for processes that are heterogeneous, path-dependent, and analytically intractable (Smaldino, 2017).

The Methods presentation organises around the seven elements of the ODD (Overview, Design concepts, Details) protocol (Grimm et al., 2010), which provides a standardised vocabulary for documenting agent-based models. *Purpose* is set by the research questions stated above; *entities, state variables, and scales* are described in the Simulation Environment and Model Architecture subsections; *process overview and scheduling* is given in Core Mechanics; *design concepts* are addressed in Scenarios and Parameter Status; *initialisation* is described in Simulation Environment; the model uses no external *input data* and is fully theoretical; and *submodels* are documented at the equation level throughout.

Simulation Environment

The simulation was implemented in Python using Mesa (Kazil et al., 2020), an open-source agent-based modelling framework. Mesa provides a standardised architecture for defining agents and model environments, a built-in data collector for recording agent- and model-level variables at each time step, and deterministic random number generation via a seeded NumPy generator, enabling exact reproduction of any run. The complete source code is publicly available at https://github.com/yuz-f/BAThesis.

Each simulation run proceeds for 300 discrete time steps. At each step, agents act in a randomly shuffled order, environmental dynamics are updated, and all outcome variables are recorded. Two scenarios are run consecutively under identical structural parameters, differing only in the distribution of initial researcher skill profiles (see Section 2.3).

Model Architecture

The model represents a stylised scientific community composed of three types of entities: *Labs*, *Researchers*, and *Scientific Models*. Labs are static institutional objects. Researchers are active agents who learn, publish, and are subject to evolutionary selection. Scientific Models are passive environmental objects that accumulate citations and decay over time.

Labs

Ten laboratories define the institutional landscape of the simulation. Each lab is characterised by a fixed *domain fingerprint* — a vector of ten values specifying the lab's characteristic skill level in each of the ten research domains. The fingerprint determines the initial skill profiles of researchers hired into the lab and serves as an institutional anchor toward which researchers' skills drift over time (see Section 2.2.5). Lab fingerprints are not updated through learning; institutional character is maintained by hiring.

At initialisation, each lab is assigned a randomly chosen *peak domain* in which its fingerprint value is drawn from a higher distribution, and nine *secondary domains* drawn from a lower distribution. The specific distribution parameters differ between the specialist and generalist scenarios (see Section 2.3).

Researchers

Fifty researchers are distributed evenly across the ten labs (five per lab). Each researcher is characterised by a personal *domain skill vector* of ten values, one per research domain, each bounded in the interval [0.01, 0.95]. Initial skills are drawn from a normal distribution centred on the researcher's home lab's fingerprint, with a small noise term ($\sigma = 0.03$), so that researchers begin with profiles close to but not identical to their institutional template.

Researchers track publications, cumulative reputation, a count of steps spent in each activity mode, and their current research domain. Reputation is the primary currency of evolutionary selection: it accumulates through successful publications and decays passively at a rate of 0.2% per step, so that recent productivity counts more than historical output.

What `domain_skill` represents — and what it does not. The skill vector is intended as a compressed, single-dimension proxy for a researcher's effective capacity to engage with work in a given domain — combining domain knowledge, methodological competence, and familiarity with that domain's prevailing techniques into one scalar. It is not a measurement of any particular real-world construct (expertise, IQ, publication count, or h-index), and no real researcher's skill profile could be directly mapped onto a vector in this model. The construct is deliberately abstract: its purpose is to make a parsimonious distinction between specialist and generalist *profile shapes* possible, while remaining agnostic about which sub-component of competence is doing the work. This is appropriate for a model that asks how profile-level differences propagate through community dynamics, but it carries a real cost: parameter recovery for individual researchers is not in principle possible, and the model is therefore not suitable for inference about individuals. It is suitable for comparing aggregate properties of populations whose skill distributions differ.

Scientific Models

A *Scientific Model* is a passive object representing a published scientific contribution within one of the ten domains. It carries four key attributes. Its *domain* identifies which research area it belongs to. Its *complexity vector* mirrors the domain fingerprint structure — a ten-element profile of how demanding the model is to work with across domains — drawn from the publishing researcher's current skill profile with noise ($\sigma = 0.05$). Its *truthfulness* ($\in [0, 1]$) measures scientific accuracy and productivity, as described below. Its *salience* ($\in [0, 1]$) measures current community attention — how visible and actively used the model is.

Truthfulness and domain caps. Each domain has a *truthfulness cap* drawn at initialisation from a uniform distribution over [0.35, 0.60], representing a ceiling on how accurate the best available model in that domain can currently be. When a new model is published, its truthfulness does not simply equal the researcher's skill; instead, it asymptotically approaches the domain cap. Specifically, a new model's truthfulness closes a fraction of the remaining gap between the current domain maximum and the cap, where the fraction is drawn from a Beta distribution whose shape parameter increases as the domain maximum grows. This produces diminishing returns: early models in an immature domain improve quickly; later models improve only incrementally. Caps grow continuously each step via asymptotic drift toward a hard ceiling of 0.95 (at rate 0.005 per step), so domains retain marginal scientific value throughout the simulation without abrupt discontinuities.

What `actual_truthfulness` represents — and what it does not. The actual-truthfulness attribute is intended to capture the probability that the model's central claim would be reproduced under independent, methodologically matched testing — its replicability under the model's own dynamics, not its correspondence to any external ground truth. It is therefore closer to a signal-to-noise quantity than to a metaphysical "truth": a high-truthfulness model is one whose findings would survive replication attempts by competent researchers; a low-truthfulness model would not. The construct is consciously a simplification. In real psychology, the distinction between a result that "is true" and one that "replicates" is contested precisely because the underlying phenomena may not be stable enough for the question to have a determinate answer (Eronen & Bringmann, 2021). The model presupposes a determinate answer in order to give the dynamics something to converge toward; the limitations of this presupposition are discussed below. The reported-truthfulness attribute, by contrast, represents the community's perception of the model's quality at any moment, and equals actual truthfulness plus an additive bias term drawn at publication time. The gap between actual and reported is the model's central representation of publication bias.

Salience dynamics. A model's salience begins at 0.5 at publication and is governed by two opposing forces. Each time another researcher successfully uses the model, it gains 0.05 salience (a citation). Each time step, salience decays at a base rate of 0.025. Crucially, high-truthfulness models resist decay as long as they are actively cited: the effective decay rate is reduced in proportion to the model's truthfulness, weighted by a *recency factor* that equals one when the model was cited in the current step and falls linearly to zero after 20 steps without a citation. A model that stops being cited therefore loses its truthfulness shield within 20 steps and then decays at the full base rate, regardless of how accurate it is. This creates era dynamics: superseded models fade on a predictable timescale and are gradually replaced by newer work. In addition, when a domain reaches high saturation (≥ 70% of its cap), all models in that domain incur an extra salience penalty each step, discouraging further investment in crowded areas.

Epistemic Landscape

Each research domain is assigned a two-dimensional *epistemic landscape* at initialisation, following the conceptual framework of Weisberg & Muldoon (2009). The landscape is a Gaussian-mixture potential-energy surface over the unit square $[0, 1]^2$, composed of *valleys* (Gaussian wells, three per domain) representing stable theoretical paradigms and *peaks* (Gaussian bumps, four per domain) representing unstable or intermediate theoretical positions. The height function $h(x, y) \in [0, 1]$ encodes epistemic instability: $h \approx 0$ at a valley bottom corresponds to well-established, broadly understood theory — the analogue of Newtonian mechanics occupying a low valley, with general relativity a higher but still well-defined attractor; $h \approx 1$ at a peak top corresponds to contested territory whose conditions are narrow, poorly characterised, and prone to revision. Stability is $s = 1 - h$.

Each Scientific Model carries a position on its domain's landscape, set at publication to the publishing researcher's current theory-space position in that domain. Researchers maintain a per-domain theory-space position that evolves through research activity: successful exploitation assimilates a researcher's position 10% toward the cited model's location, implementing slow convergence toward productive theory-space. During exploration, new models are placed near the researcher's current position with Gaussian noise whose standard deviation shrinks with experience — from $\sigma = 0.20$ for a researcher's first paper in a domain to $\sigma = 0.04$ after sixteen publications — so that novices enter from scattered positions while experts cluster in a tightening region. Follow-up models (derivative publications triggered by successful exploitation) are placed one gradient-descent step closer to the nearest valley, so that mature research programmes converge on stable theoretical anchors cumulatively.

Figure 2 shows one domain's epistemic landscape together with the spatial distribution of models that the two scenarios produce by step 300. Panels A and B render the height function

$h(x, y)$ as a 3D surface and a top-down heatmap, respectively, with the seed model marked. Panels C and D show, for the same domain, the position of every model published over 300 steps under each scenario, with currently-active models highlighted. The visual contrast between Panels C and D previews the quantitative results that follow: specialist research programmes produce active literatures concentrated in the valley region (left), while generalist scenarios produce a more dispersed active literature. The publication-history background (faint dots) shows that both scenarios *publish* across the full theory-space, but only valley-positioned work *survives* as the actively cited literature.

Landscape stability enters five downstream quantities. *Replication probability* (Equation 1) is scaled by a stability factor $(0.75 + 0.25\, s)$: valley models replicate more easily because their theoretical conditions are well-characterised and reproducible. *Debunking vulnerability* (Equation 5) is scaled by an instability boost $(1 + 0.5\,(1 - s))$: peak models are more susceptible to successful challenge because the conditions under which they hold are narrow and poorly understood. *Salience decay* is slowed by up to 25% for valley models (effective_decay $\times$ $(1 - 0.25\, s)$), reflecting that well-established results remain salient without constant new citations. *Publication bias inflation* receives an additional term of up to 0.15 proportional to peak height, capturing the tendency to over-report results from exciting but theoretically unstable territory. Finally, *action-selection attractiveness* is scaled by a stability-attraction factor $(1+0.30\, s)$ in the exploit value calculation: valley models look more rewarding to work on, so researchers are pulled toward stable theoretical anchors not only through follow-up dynamics but through their primary action choice. Together, these five mechanisms make the epistemic landscape a structural rather than purely cosmetic addition: stability shapes what gets done, what survives, and what gets challenged.

Core Mechanics: Exploit, Invest, Explore, and Debunk

At each step, a researcher either continues an ongoing multi-step exploitation task or selects a new action. The four possible activities are described below.

Exploit. A researcher commits to implementing an existing Scientific Model. Implementation requires multiple consecutive steps, reflecting that engaging seriously with a published model takes sustained effort. The number of steps required scales with the model's complexity in its primary domain and the domain's current saturation, and is reduced by the researcher's own skill in that domain. When the required steps have been completed, the researcher makes an implementation attempt.

The probability of a successful attempt combines two factors: the researcher's raw proficiency in the model's domain, and a *similarity* term that measures how well the researcher's full skill profile matches the model's complexity profile, quantified as the Pearson correlation between the two vectors, floored at zero and weighted by domain proficiency. Formally, the similarity of researcher i to model m is

$$\text{sim}_{i,m} = \text{proficiency}_i(m) \times \max\big(0,\ \text{Cor}(\mathbf{s}_i,\ \mathbf{c}_m)\big)$$

where $\mathbf{s}_i$ is the researcher's skill vector and $\mathbf{c}_m$ is the model's complexity vector. The overall success probability is then

$$P(\text{success}) = \Big(\text{sim}_{i,m} + \bar{s}_i\,(1 - \text{sim}_{i,m})\Big) \times \frac{\theta_{\text{actual}}}{\theta_{\text{reported}}} \times \underbrace{(0.75 + 0.25\,s_m)}_{\text{stability factor}} \qquad (1)$$

where $\bar{s}_i$ is the researcher's mean skill across all domains, $\theta_{\text{actual}}/\theta_{\text{reported}}$ is the truth ratio, and s_m is the model's landscape stability (see Section 2.2.4). The truth ratio is the key mechanism through which publication bias enters replication dynamics: a model reported 20% above its actual quality is approximately 20% less likely to replicate successfully, regardless of researcher skill. Systematic inflation of reported effect sizes in published literature — particularly pronounced in psychology — is well-documented empirically (Fanelli, 2010). A researcher who is skilled in the right domain and whose broader expertise aligns with the model's demands is most likely to succeed. The mean-skill floor ensures that any researcher with genuine competence always has a non-negligible success probability. The implementation in `researcher.py` reads directly:

```
def success_probability(self, m):
    sim = self._similarity(m)           # proficiency × max(0,
Cor(skills, complexity))
    avg = float(np.mean(self.domain_skills))
    base = sim + avg * (1.0 - sim)
    truth_ratio      = m.actual_truthfulness /
max(m.reported_truthfulness, 1e-8)
    stability_factor = 0.75 + 0.25 * m.landscape_stability  # valley
models replicate more easily
    return float(np.clip(base * truth_ratio * stability_factor, 0.0,
    1.0))
```

On success, the researcher earns one publication, gains reputation proportional to the model's current *reported* truthfulness and salience (agents are rewarded based on what the community perceives, not the hidden actual quality), cites the model (raising its salience), and assimilates toward the model's complexity profile (see below). With a small probability that decreases as the domain fills up, the researcher also spawns a follow-up model, representing the opportunistic extension of successful work.

On failure, two additional processes are triggered based on the researcher's domain proficiency. First, an *expert truth correction* mechanism operates when proficiency exceeds 0.50: an expert who fails to replicate a result has enough domain knowledge to attribute the failure to the model's actual limitations rather than their own incompetence. Each such failure erodes the model's actual truthfulness by a small amount,

$$\Delta\theta_{\text{actual}} = -\,\text{proficiency} \times 0.015 \times \left(1 + \underbrace{(\theta_{\text{reported}} - \theta_{\text{actual}})}_{\text{bias gap}}\right) \tag{2}$$

so that models with inflated reported quality erode faster. This creates a continuous, low-amplitude truth-revelation process distinct from the large discrete correction of a formal debunk. The implementation (`researcher.py`, failure branch of `_attempt`):

```
proficiency = self.domain_skills[target.domain]
if proficiency > 0.50:
    gap_inflation = target.reported_truthfulness -
target.actual_truthfulness
    correction    = proficiency * 0.015 * (1.0 + gap_inflation)
    target.actual_truthfulness = max(0.01, target.actual_truthfulness -
correction)
```

Second, a researcher with domain proficiency exceeding 0.35 has a probability proportional to their proficiency of initiating a formal debunk — the intuition being that an expert who cannot reproduce a result is well-positioned to mount a systematic challenge.

Invest. When a model's complexity in its primary domain exceeds the researcher's skill by more than a fixed training threshold, the researcher cannot yet attempt exploitation and instead invests in training. One training step moves the researcher's skills toward the model's complexity profile via the assimilation mechanism (see below), at a rate scaled by how well the researcher's overall profile already aligns with the model's demands.

Explore. The researcher publishes a new Scientific Model in a domain they select themselves. The value of exploring domain d is

$$V(\text{explore},\ d) = \frac{\text{skill}_d \times \max(0,\ 1 - \text{saturation}_d)}{1 + \ln(1 + n_d) + \text{own\_pubs}_d} \tag{3}$$

where n_d is the current number of models in domain d and own_pubs_d is the researcher's own prior publication count there. Saturated or personally exhausted domains offer little marginal value; fresh or underdeveloped domains are attractive targets.

A key extension to the basic explore action is the **breakthrough mechanic**. With a small probability that scales with the researcher's squared domain skill, the published model constitutes a paradigm-shifting breakthrough:

$$P(\text{breakthrough}) = \text{skill}_d^2 \times 0.10 \tag{4}$$

At skill 0.55 this yields approximately a 3% breakthrough rate; at skill 0.90 it is approximately 8%. A breakthrough model receives higher initial truthfulness (drawn from a Beta distribution concentrated near the top of the remaining gap-to-cap), starts with a salience of 0.90 rather than the standard 0.50, and triggers a *salience shock*: all incumbent models in the domain lose 65% of their current salience as the community's attention pivots to the new result.

This creates era dynamics — trend cycles where a dominant paradigm is periodically displaced rather than accumulating indefinitely. The implementation (`researcher.py`, `_explore`):

```
skill = self.domain_skills[domain]
is_breakthrough = self.random.random() < skill ** 2 * 0.10

self.model.spawn_model(..., breakthrough=is_breakthrough)

if is_breakthrough:                            # salience shock to
incumbents
    new_uid = self.model.scientific_models[-1].uid
    for m in self.model.scientific_models:
        if m.domain == domain and m.uid != new_uid:
            m.salience = max(0.0, m.salience * 0.35)
```

Debunk. A researcher attempts to disprove an existing model. Debunking is not a freely chosen action; it is triggered as a byproduct of failed exploitation by a researcher with domain proficiency above 0.35. The probability of a successful debunk is

$$P(\text{debunk success}) = \text{proficiency} \times (1 - \theta_{\text{actual}}) \times \underbrace{(1 + 0.5\,(1 - s_m))}_{\text{instability boost}} \tag{5}$$

Experts are therefore most likely to expose weak models, and genuinely high-quality models are largely immune even when a very skilled researcher fails to replicate them. Models positioned near landscape peaks are additionally vulnerable: the instability boost scales from

1.0 (valley) to 1.5 (full peak), reflecting that results from theoretically contested positions are more susceptible to challenge. A successful debunk reduces the target model's truthfulness by 25% and its salience by 0.15, cascades partial damage to directly derivative models (truthfulness $\times 0.88$ and salience -0.08 for children), and earns the researcher a publication and reputation proportional to the impact of the discredited model. The original author incurs a reputational penalty equal to half of the debunker's reward. The implementation (`researcher.py`, `_debunk`):

```
proficiency       = self.domain_skills[target.domain]
instability_boost = 1.0 + 0.5 * (1.0 - target.landscape_stability)  #
peak models more vulnerable
success_prob      = proficiency * (1.0 - target.actual_truthfulness) *
instability_boost
if self.random.random() < success_prob:
    target.actual_truthfulness = max(0.01, target.actual_truthfulness *
0.75)
    target.salience            = max(0.0,  target.salience - 0.15)
    # cascade to derivative models
    for m in self.model.scientific_models:
        if m.parent_uid == target.uid:
            m.actual_truthfulness = max(0.01, m.actual_truthfulness *
0.88)
```

Action selection. At the start of each step (if the researcher is not mid-exploitation), available models are partitioned into *exploit candidates* (complexity gap $\leq$ training threshold) and *invest candidates* (gap > threshold). The top two models by expected value are retained, and an explore option is always added — directly instantiating the explore–exploit tradeoff that March (1991) identifies as the central tension in adaptive systems. The researcher then selects probabilistically among these options. The selection weight for each candidate is scaled by a *skill bias* term,

$$\text{skill\_bias}_d = \left(\frac{\text{skill}_d}{\bar{s}}\right)^{0.5} \tag{6}$$

where $\bar{s}$ is the researcher's mean skill. This is a parameter-free mechanism: a specialist with a skill ratio of ≈ 2 in their peak domain receives a weight multiplier of ≈ 1.41 for work in that domain, while a generalist with near-uniform skills receives a multiplier close to 1.0 everywhere. The bias emerges naturally from the skill distribution rather than being hardcoded, so it strengthens as specialisation deepens through assimilation. This ensures that both the intrinsic quality of a model and its current community visibility influence the researcher's choice, without making behaviour deterministic.

Assimilation. Working on a model gradually shifts the researcher's skills toward the model's complexity profile. The skill update follows an S-curve:

$$\Delta\text{skill}_d = r \times \text{skill}_d \times (1 - \text{skill}_d) \times 4 \times \max(0,\ c_d - \text{skill}_d) \times f_d \tag{7}$$

where r is the base learning rate (0.06 for attempts, 0.08 for training), c_d is the model's complexity in domain d, and f_d is a focus weight ($f_d = 1$ for the model's primary domain, $f_d = 0.12$ for all other domains). The gain is largest when current skill is near 0.5 and approaches zero at the extremes (no foundation to build on, or diminishing returns near mastery) —

a logistic form consistent with empirical accounts of staged skill acquisition (Anderson, 1982). Learning is focused on the model's primary domain; secondary domains receive only 12% of the primary learning rate, preventing general engagement from silently inflating every skill. Skills only move upward through assimilation — working on a difficult model cannot reduce existing competence.

Fingerprint drift. To represent the pull of institutional norms — the tendency of researchers to converge toward the methods and problems valued by their immediate community (Crane, 1972) — each researcher's non-peak skills drift passively toward their home lab's fingerprint every step. Skills above the fingerprint baseline decay at 0.2% per step; skills below it are pulled upward at 0.1% per step. The researcher's peak domain is exempt, allowing genuine specialisation to grow freely.

Realism Extensions

Five mechanisms extend the core model toward empirically documented features of scientific communities.

Career stages. A `career_age` counter increments each step and resets to zero on researcher replacement. The explore action weight is multiplied by $(1 + \max(0,\ 1 - t/150))$, where t is career age, doubling exploratory tendency at career start and fading to the base weight by step 150. This captures the documented shift from broad exploration toward consolidation over the course of an academic career: early-career researchers invest more heavily in new domains, while senior researchers increasingly exploit established competencies (Petersen et al., 2011).

Social learning. Each lab maintains an exponentially decayed count of recent domain successes (decay factor 0.90 per step, half-life ≈ 6 steps). This signal enters the exploit action value as a multiplicative bonus $1 + \lambda \cdot \tanh(\text{signal})$ with $\lambda = 0.30$, pulling lab-mates toward recently productive areas. The mechanism captures the invisible-college effect: informal success signals within a research group concentrate collective attention on shared domains and methods (Crane, 1972).

Competitive pressure and publication inflation. At spawn time, the bias draw mean shifts upward by up to 0.08 proportionally to how far the publishing researcher's reputation falls below the current field median. Researchers under career pressure therefore inflate reported quality more than successful ones — a dynamic consistent with evidence that competitive publication pressure drives methodological sloppiness and result inflation (Smaldino & McElreath, 2016).

Misconduct pathway. With a base probability of 0.05 that scales with competitive pressure (up to 0.15 at maximum pressure), a publication's bias is drawn from $\mathcal{N}(0.22,\ 0.06)$ rather than the standard distribution, representing strategic rather than incidental inflation. Fang et al. (2012) find that 67% of retracted biomedical articles involved misconduct; the misconduct pathway ensures that a structurally fragile subset of the literature exists independently of researcher skill.

Matthew effect on opportunities. The exploit action value includes a salience amplification factor $1 + 0.30 \cdot \tanh(\rho \times 0.05)$, where ρ is the researcher's current reputation. Elite researchers are therefore drawn more strongly toward the most prominent work in each domain, capturing the observation that prior success compounds access to high-quality research opportunities (Merton, 1968).

Evolutionary Selection and Lab Turnover

Every 40 steps, an evolutionary selection event occurs. Researchers are ranked by the reputation they have accumulated since the previous selection event — recent performance, not career totals. The bottom 25% are removed and replaced by mutated copies of elite researchers (the top

25% by the same criterion). Mutation perturbs each skill in the child's profile by a small amount ($\sigma = 0.04$), introducing variation while preserving the general shape of high-performing profiles. Each replacement researcher joins the same lab slot as the removed researcher, so the population size and lab distribution remain constant.

Following each selection event, the model evaluates whether any lab's average recent performance falls below a threshold relative to the field average. If the weakest lab's mean per-researcher reputation gain is less than 50% of the overall mean, that lab's fingerprint is replaced with a newly drawn random fingerprint. Current researchers in the lab keep their personal skills but now drift toward the new institutional direction, representing a change in the lab's research identity. This mechanism introduces institutional-level turnover without altering the total number of labs or researchers.

Scenarios: Specialist vs. Generalist

The simulation is run under two contrasting parameter configurations that differ solely in the distribution from which non-peak domain skills in lab fingerprints are drawn. All structural parameters — number of labs, researchers, domains, selection interval, and so on — are identical across scenarios.

In the **Specialist** scenario, secondary domain skills are drawn from a low distribution (mean = 0.25, SD = 0.06), producing researchers whose skill profile has one pronounced peak and a flat, weak baseline elsewhere. Labs are deeply specialised in their assigned domain and have little capacity to engage with others.

In the **Generalist** scenario, secondary domain skills are drawn from a higher distribution (mean = 0.33, SD = 0.08), producing researchers with a moderate primary peak but a substantially elevated baseline across all other domains. Labs retain a primary focus but can engage meaningfully with neighbouring research areas.

In both scenarios the peak domain is drawn from the same distribution (mean = 0.55, SD = 0.07–0.08). The key manipulation is therefore the *gap* between peak and non-peak skill, not the absolute level of peak expertise.

A third **Uniform** scenario is run as a control. Here every researcher's skill in every domain is drawn from a single low-variance distribution centred at 0.40 (SD = 0.02 across all domains, with no peak/non-peak distinction). Researchers in this scenario have nearly identical, flat profiles, with no preferred domain. The Uniform scenario serves as a falsification baseline for H1 and H2: if the directional results in those hypotheses simply reflect any non-trivial population dynamic in the model, they should appear in the Uniform scenario as well. If, however, they require skill *heterogeneity* between researchers — the structural feature manipulated by the Specialist/Generalist contrast — they should be absent or substantially attenuated under Uniform skills. The Uniform scenario therefore tests whether the headline findings are a function of the manipulation or of the model's dynamics more generally.

Table 1
Fixed and varied simulation parameters across the three scenarios. The Uniform scenario sets peak and non-peak parameters equal so that all researchers have flat skill profiles centred near 0.40, with no preferred domain.

Parameter	Specialist	Generalist	Uniform
Peak domain skill mean	0.55	0.55	0.40
Peak domain skill SD	0.07	0.08	0.02
Non-peak domain skill mean	0.25	0.33	0.40
Non-peak domain skill SD	0.06	0.08	0.02
Number of labs	10	10	10
Researchers per lab	5	5	5
Number of domains	10	10	10

Parameter	Specialist	Generalist	Uniform
Simulation steps	300	300	300
Selection interval (steps)	40	40	40
Cull fraction	0.25	0.25	0.25
Mutation SD	0.04	0.04	0.04
Training threshold	0.30	0.30	0.30

Parameter Status: Empirical, Heuristic, and Structural Choices

The model contains roughly thirty free parameters in addition to the manipulation variables shown in Table 1. Their status is mixed, and the thesis benefits from making this explicit rather than treating them as a uniform list. Following the categorisation framework familiar from agent-based-modelling methodology, the parameters fall into four classes.

Empirically motivated parameters are anchored, however loosely, to the published literature. The misconduct base rate (0.05 per publication) is connected to Fang et al. (2012)'s report that 67% of retracted biomedical articles involved misconduct, which sets a rough order-of-magnitude floor. The career-stage decay horizon (150 steps) is consistent with Petersen et al. (2011)'s empirical work on early-career risk-taking and consolidation timescales. The OSC failure-rate empirical regime (Open Science Collaboration, 2015) is referenced post-hoc as a sanity check on the simulated failure rate, not as a calibration target. These are the only parameters with non-trivial empirical anchoring; everything else is theoretically or structurally motivated.

Theoretically motivated parameters are chosen on the basis of formal arguments rather than data. The skill-bias exponent (0.5) is a square-root dampener that prevents extreme specialists from monopolising action selection; it follows the logic that competitive advantage should grow more slowly than raw skill. The S-curve assimilation rule (Anderson, 1982) uses a logistic gain function appropriate for skill acquisition. The breakthrough probability ($\text{skill}^2 \times 0.10$) embeds the assumption that breakthroughs are quadratically rare in skill — a working assumption rather than a measured rate. The epistemic-landscape parameters (3 valleys, 4 peaks per domain, Gaussian widths in [0.03, 0.10] for valleys and [0.02, 0.08] for peaks) are inherited from the Weisberg & Muldoon (2009) framework as theoretical scaffolding.

Heuristic parameters are chosen to produce numerically stable, non-degenerate dynamics over realistic horizons. Examples include the selection interval (40 steps), the cull fraction (0.25), the mutation standard deviation (0.04), the salience-decay base rate (0.025), the cap-growth rate (0.005), the reputation-decay multiplier (0.998), and the various stability-factor magnitudes (0.75–1.30 range). These were selected during development to ensure that the system reaches quasi-steady-state within 300 steps without collapsing or saturating. They are defensible but not derived; alternative values within an order of magnitude would produce qualitatively similar dynamics, as the sensitivity analysis confirms for the parameters tested.

Convenience parameters are arbitrary scale or bounding choices that should not affect substantive conclusions. Skill values are bounded to [0.01, 0.95] to avoid numerical edge cases; the truthfulness ceiling (0.95) prevents asymptotic divergence; initial salience (0.50) sets a neutral starting point; the simulation horizon (300 steps) is chosen for tractability and is verified to exceed the time required for differentiation between scenarios (Section 3.2).

This classification clarifies what kinds of objection different parameters are vulnerable to. Empirical parameters can be challenged by new empirical evidence; theoretical parameters by alternative theoretical commitments; heuristic parameters by demonstrating that alternative choices change the conclusions; convenience parameters by showing that the bounds bind on substantive results. The sensitivity analysis (Section 3.8) addresses the heuristic class for six of the most consequential parameters; the layer-decomposition ablation (Section 3.7) addresses the theoretical class by quantifying the contribution of each architectural layer. These are the relevant defences. The parameters that most call out for further investigation are the heuristic ones not currently included in the sensitivity sweep — a limitation acknowledged here and recommended for future work.

Model Evaluation Criteria

Constructing a simulation model involves navigating a fundamental tension between fidelity and comprehensibility. As Bonini's paradox states, a model that perfectly mirrors the complexity of its target system becomes as difficult to understand as that system itself (Smaldino, 2017). The purpose of a model is not to replicate reality in full but to isolate the mechanisms sufficient to generate the phenomenon of interest, with only as many free parameters as the question demands.

The present model is evaluated against three criteria. First, *parsimony*: each parameter and mechanism must earn its place by contributing to the qualitative pattern of outcomes. Parameters that can be removed without changing the model's behaviour were not included. Second, *transparency*: all assumptions are explicit and quantified, so that any reader can re-run the simulation, verify the reported outcomes, and test alternative parameter settings. Third, *discriminability*: the two scenarios must produce qualitatively distinct patterns of behaviour, so that the comparison yields interpretable conclusions rather than a null result. Whether the model meets these criteria is evaluated in the Results and Discussion sections.

Statistical power and confidence intervals. With $n = 30$ independent seeds per scenario, the minimum effect size detectable by a two-sample Mann-Whitney U test at $\alpha = .05$ (two-sided) and power $1 - \beta = .80$ is approximately $d = 0.74$. The principal H1 effect ($d = 0.78$) sits just above this threshold; H2 ($d = 2.14$) is far above it. For the layer-decomposition ablation (Section *Layer-Decomposition Ablation*; $n = 15$ per cell), the minimum detectable d rises to approximately 1.05; the L1 null result is therefore reported as *consistent with* but not *demonstrative of* the absence of an effect at base architecture, with the substantive ablation inference resting on the *change* in effect size as layers are added rather than on the absolute null at L1. All 95% confidence intervals reported are computed from the seed-level distribution of the metric using Student's t with $n - 1$ degrees of freedom; the seed-level distributions are approximately normal at $n = 30$, and a t-based interval is the standard analytic choice in this regime. Holm-Bonferroni correction across the four-test confirmatory-and-exploratory family (H1, H2, H3, H4) leaves H1 ($p = .006$) and H2 ($p < .0001$) significant; H3 ($p = .684$) and H4 ($p = .165$) do not survive correction, consistent with their already failing at uncorrected α. Ablation tests and Uniform-control comparisons are reported with uncorrected α and treated as exploratory extensions of the confirmatory family.

Results

Each scenario was run under 30 independent random seeds, producing 30 complete 300-step simulations per condition. All inferential tests are two-sided Mann-Whitney U tests, and effect sizes are reported as Cohen's d. The confirmatory threshold was set at $\alpha = .05$; the two prespecified hypotheses (H1, H2) are evaluated at this level. The two exploratory hypotheses (H3, H4) are treated as hypothesis-generating and interpreted without formal correction.

Confirmatory Analyses

H1: Replication failure rate. The specialist scenario produced a mean replication failure rate of .62 (95% CI [.60, .63]) compared to .59 (95% CI [.58, .60]) in the generalist scenario. The difference was statistically significant: $U = 638.0$, $p = .006$, $d = 0.78$. H1 is supported. The direction is as predicted: specialist researchers, whose skill profiles are poorly matched to models outside their peak domain, fail to replicate at a higher rate than generalist researchers who maintain a meaningful baseline across all domains.

The simulated specialist failure rate (62%) is in the same range as the non-replication rate of 61% reported by Open Science Collaboration (2015) in their large-scale empirical audit of psychological findings. The two quantities are not directly comparable. The OSC figure is the proportion of significant published findings that did not produce a significant result under a matched-method replication attempt; the simulated rate is the proportion of multi-step exploitation tasks that did not succeed under the model's success-probability rule. The constructs differ in what counts as a failure, what counts as an attempt, and what counts as a study. The numerical proximity is therefore suggestive rather than calibrational — no parameter was tuned to it — but it indicates that the model produces failure rates in an empirically plausible regime rather than a degenerate one. Reading anything stronger into the match would over-interpret a single point comparison.

H2: Domain concentration of publications. The Gini coefficient of domain-level publication counts was .139 (95% CI [.131, .147]) in the specialist scenario and .093 (95% CI [.086, .101]) in the generalist scenario. This difference was highly significant and very large: $U = 848.0$, $p < .0001$, $d = 2.14$. H2 is supported. The publication landscape in the specialist scenario is substantially more concentrated: a few domains attract disproportionate publication activity while others are comparatively neglected. In the generalist scenario, the skill-bias term (Equation 6) provides only a weak pull toward any single domain, resulting in a more even distribution of research effort. The effect size for H2 is larger than for H1 ($d = 2.14$ vs. $d = 0.78$), indicating that domain concentration is the more reliable structural signature of skill specialisation.

Temporal Dynamics

The end-state metrics reported above summarise 300-step trajectories. To show how the H1 and H2 differences emerge over time, Figure 4 plots both metrics across the full simulation horizon for all three scenarios, with mean trajectories and 95% confidence bands across 30 seeds.

Both metrics differentiate the scenarios within the first 50–75 steps and remain stable for the remaining 225 steps. For H1, all three scenarios begin at high failure rates ($\approx .68$ at step 1) reflecting the small initial pool of replication attempts; failure rates then decrease as the literature accumulates and researchers find better-matched models. The Generalist trajectory descends fastest and stabilises lowest; Specialist and Uniform trajectories descend more slowly and stabilise above. For H2, the Gini coefficient starts high ($\approx .20$) reflecting the seed-only domain configuration in the early steps and then descends sharply as researchers begin publishing across all ten domains. By step 50 the three scenarios have separated into their final ordering — SPEC > GEN > UNIFORM — and remain within their respective bands for the rest of the simulation.

The temporal stability of the differences is methodologically reassuring: the confirmatory results in Section 3.1 do not depend on a particular endpoint choice. Reporting the same statistics at step 100, 200, or 300 would give qualitatively the same answer.

Exploratory Analyses

H3: Mean researcher reputation. Mean end-of-simulation reputation was 10.80 (95% CI [10.41, 11.20]) in the specialist scenario and 10.79 (95% CI [10.41, 11.18]) in the generalist scenario. The difference was not significant: $U = 478.0$, $p = .684$, $d = 0.008$. H3 is not supported.

The null result is interpretable in light of the action distribution data. Specialist researchers spent 56% of steps in training, compared to 43% for generalists; generalists spent 33% of steps in successful exploitation, compared to 20% for specialists. These are structurally different paths to the same career-level outcome. Specialists produce fewer successful replications per unit time but earn more per success when they do exploit, because they concentrate in high-salience, high-truthfulness models in their peak domain, which also tend toward stable valley positions in the epistemic landscape. Generalists accumulate more successful replications across a broader territory but earn less per event on average. The null result does not indicate that community structure is irrelevant to how researchers work; it indicates that overall reputation level is not a discriminating outcome, while the structure of how that reputation is achieved differs substantially.

H4: Debunking vulnerability. The debunk impact rate — the fraction of career reputation earnings permanently lost to debunking events — was .00040 (95% CI [.00024, .00060]) in the specialist scenario and .00080 (95% CI [.00030, .00130]) in the generalist scenario. The difference did not reach statistical significance: $U = 356.0$, $p = .165$, $d = -0.40$. H4 is not supported. The direction is the *reverse* of the pre-study hypothesis — generalists lost more reputation to debunking, not specialists — consistent with findings from the earlier model iteration, but the effect falls short of the significance threshold and carries wide confidence intervals in the generalist condition.

The structural mechanism underlying this directional pattern is informative even in the absence of significance. Specialist researchers publish primarily in their peak domain, where their high skill produces models with above-average actual truthfulness. Equation 5 shows that debunk success probability scales with $(1 - \theta_{\text{actual}})$ and is further amplified for models in unstable theory-space positions: specialist research programmes, which converge toward valley positions via gradient descent, benefit from both high truthfulness and high landscape stability, making their output doubly resistant to challenge. Generalist researchers publish across multiple domains at intermediate skill levels, producing models with lower average truthfulness that are exposed at less stable landscape positions. The introduction of the epistemic landscape appears to have increased variance in the debunking process — the instability boost raises debunking rates generally, compressing the proportional difference between scenarios and widening confidence intervals — which accounts for the loss of significance relative to the earlier model. The direction of the effect, however, is robust across both model versions.

Supplementary: Mechanism Check

The action-distribution data confirm that the model's internal mechanics are behaving as expected. Across all steps pooled within each scenario, exploit steps constitute 19.5% of activity for specialists and 33.2% for generalists; training steps constitute 56.4% and 43.1% respectively; explore steps are nearly equal (24.1% vs. 23.6%). The elevated training burden for specialists directly reflects the model's training threshold mechanism: specialists frequently encounter models whose complexity in secondary domains exceeds their skill, triggering sustained training investment that does not translate into publications. This time cost is the proximal cause of the lower exploitation rate, which in turn limits the diversity of models that specialist researchers engage with and replicate. The higher explore fraction compared to earlier model iterations (≈24% vs. ≈19%) reflects the epistemic landscape's effect on exploration value: novel theory-space positions are more attractive when the gradient toward valleys draws researchers toward productive territory.

Mean actual truthfulness converged to identical values in both scenarios (.842 vs. .842, $d = .012$, $p = .982$), as did the bias gap (.108 vs. .108, $d = -.012$, $p = .982$). Both scenarios drive the literature toward the same aggregate knowledge state; the community structure affects *who* *fails* along the way, not *where the domain ends up*. The reported truthfulness ceiling at .95 reflects both scenarios saturating their domains' truthfulness caps by step 300, consistent with the cap growth mechanism described in the Methods section.

Control Check: Uniform Skill Distribution

To distinguish whether the SPEC > GEN findings for H1 and H2 require skill *heterogeneity* — the structural feature manipulated by the Specialist/Generalist contrast — or simply reflect any non-trivial population dynamic in the model, a third *Uniform* scenario was run with all researchers having near-identical flat skill profiles centred at 0.40 (Section 2.3). The control was conducted under 30 independent seeds matching the main experiment. It yields four findings, two of which sharpen the headline interpretation and two of which complicate it productively.

H2 is cleanly confirmed: domain concentration requires skill heterogeneity. The Gini coefficient under Uniform skills was .075 (95% CI [.066, .084]), substantially lower than both the Specialist (.139) and Generalist (.093) scenarios. The full ordering is SPEC > GEN > UNIFORM, with all pairwise comparisons highly significant (SPEC vs UNIFORM: $U = 875$, $p < .0001$, $d = 2.79$; GEN vs UNIFORM: $U = 658$, $p = .002$, $d = 0.82$). With flat profiles, the skill-bias term in action selection (Equation 6) provides no preferential pull toward any domain, and the Gini falls to a near-baseline level driven only by stochastic variation in publishing activity. This converts H2 from a directional finding to a calibrated one: domain concentration emerges not because the model has any non-trivial dynamic, but specifically because skill heterogeneity is present, and the magnitude of concentration scales with the degree of heterogeneity.

H1 is more nuanced than the binary contrast suggested. The Uniform failure rate (.645, 95% CI [.639, .650]) was not lower than either the Specialist or Generalist rate but *higher* than both, and the differences are large (SPEC vs UNIFORM: $U = 209$, $p < .001$, $d = -1.02$; GEN vs UNIFORM: $U = 19$, $p < .0001$, $d = -2.61$). The relationship between skill specialisation and replication failure is therefore not monotonic: a community of uniformly mediocre researchers produces even higher failure rates than a community of specialists working partly in mismatched domains. The action-fraction data clarify why. Uniform researchers spent 91% of steps in exploitation (compared to 19% for Specialists and 33% for Generalists), because a flat skill of 0.40 sits just above the training threshold (0.30) for most models, allowing direct attempts without sustained training. The combined effect of high attempt rate and mediocre proficiency is many failed replications. This finding is methodologically important: it indicates that the failure-rate effect is not driven by specialisation per se but by the interaction between *who is qualified to attempt what* and *how often that interaction happens*. The Specialist and Generalist scenarios both restrict attempts (through training requirements) in ways that the Uniform scenario does not, and this restriction protects against the worst replication outcomes despite producing higher failure rates *among the attempts that are made*.

The H3 null is explained by equal peak skill, not equivalent dynamics. Uniform researchers earned approximately half the reputation of either Specialists or Generalists (5.61 vs ~10.80; $d = 5.8$ in both pairwise comparisons). This makes clear what the SPEC vs GEN null result was hiding: the two scenarios share an identical peak-skill distribution (mean = 0.55), and reputation accumulation in the model is bounded by the highest-quality work a researcher can produce. The absolute ceiling on skill, not the shape of the distribution beneath it, determines the magnitude of reputation; the SPEC/GEN null is therefore not a statement that skill structure does not matter for careers — it is a statement that careers in this model are paced by peak ability, while *how* researchers build toward that peak (via narrow specialisation or broad coverage) differs structurally without affecting the aggregate level.

H4's directional finding is strengthened. Uniform models were the most vulnerable to debunking (impact rate .0025, vs. .0004 for SPEC and .0008 for GEN; $d = -2.46$ vs SPEC,

$d = -1.38$ vs GEN, both $p < .0001$). The full ordering — SPEC < GEN < UNIFORM — corroborates the structural-sheltering argument advanced in the H4 Discussion: high-quality work is harder to invalidate, and Uniform researchers produce work whose mediocrity in every domain combines with elevated debunking opportunities (since any researcher can debunk in any domain) to produce the highest vulnerability. The reversal observed in the SPEC vs GEN comparison is therefore one segment of a monotonic SPEC < GEN < UNIFORM ordering, which provides a more confident structural claim than the SPEC vs GEN comparison alone supported.

Together, the Uniform control transforms the contribution of the work. H2 is sharpened (heterogeneity is the necessary condition); H1 is reframed as a U-shape rather than a monotonic story; H3 is explained as a ceiling effect; H4 is strengthened by the addition of a third ordering point. The simulation now answers questions about skill *distribution shape* rather than only about the binary specialist/generalist contrast.

Validation: Parameter Recovery and Empirical Comparison

Two further analyses test the model's methodological standing beyond the sensitivity check below. The first is internal: does the simulation output carry information about its own input parameter? The second is external: does the model produce inequality patterns at the same magnitude as documented in real academic publication distributions? Figure 6 reports both.

Parameter recovery. A grid of six values of the manipulated parameter `other_skill_mean` (0.20, 0.25, 0.30, 0.33, 0.38, 0.43, spanning the Specialist and Generalist conditions and extrapolating both directions) was crossed with 15 random seeds, yielding 90 simulations. From eight outcome metrics per run (replication failure rate, per-domain Gini, per-researcher Gini, mean publications, mean reputation, and the three action fractions), a leave-one-out cross-validated linear regression was fit to predict the underlying parameter. Recovery was strong: $R^2 = 0.96$, RMSE = 0.016, Pearson $r = 0.98$ ($p < .0001$). The simulation outputs therefore carry essentially complete information about the manipulated input parameter — a partial-identifiability check that does not establish full identifiability of the model's many other parameters but does demonstrate that the headline manipulation is recoverable from the observable outputs alone. This is a baseline methodological standard rather than a strong claim, but it rules out the possibility that the H1 and H2 results are sensitive to seeds without being driven by the manipulation.

Empirical comparison: per-researcher publication inequality. Across all 90 runs, the Gini coefficient of per-researcher publication counts at simulation end was tightly distributed around $M = 0.41$ (SD = 0.03). The empirical literature on academic productivity reports Gini coefficients of approximately 0.5 to 0.7 for publication counts across active researchers (Allison & Stewart, 1974; Nielsen & Andersen, 2021), with Nielsen & Andersen (2021) estimating annual citation Gini at 0.65–0.70 in modern data. The model therefore produces qualitatively similar inequality (substantial Gini > 0, with publishing concentrated in a minority of researchers) but at a magnitude below the empirical band. None of the 90 runs fell within [0.50, 0.70].

The undershoot is interpretable rather than disqualifying. Three structural features of the model bound how much inequality it can generate. First, the cull-and-mutate selection rule replaces the bottom 25% of researchers every 40 steps, removing the long left tail of zero-publication careers that contributes substantially to real-world Gini. Second, the simulation runs for 300 steps; real publication-Gini estimates aggregate over multi-decade career horizons during which cumulative-advantage dynamics have far longer to operate. Third, the model lacks the heavy-tailed mechanisms (extreme breakthrough cascades, persistent reputation accumulation without decay, network-driven citation amplification) that produce the most unequal publication distributions empirically. The model therefore produces *cumulative-advantage-flavored* inequality in a *bounded* form, and its 0.41 Gini reflects the shape of the dynamics that are present (skill-bias, salience, reputation) without the tail-driving mechanisms that are absent. A model that exactly reproduced 0.65 would be evidence of fitting; a model that reproduces qualitative inequality at a lower magnitude with an interpretable boundary is evidence of partial empirical correspondence with known limits.

Together these analyses sharpen the thesis's methodological standing: the manipulation is recoverable from outcomes (so the H1/H2 results reflect the manipulation, not seed noise), and the model's empirical correspondence is partial and quantitatively bounded rather than implicitly assumed. The Gini undershoot is itself an interpretable claim about which empirical features the present model architecture can and cannot reproduce.

Layer-Decomposition Ablation

A central concern with computational models of this kind is that headline findings can be implicit consequences of architectural choices made when the model was constructed, rather than substantive properties of the system being modelled. To address this concern directly, three nested model layers were run under both the Specialist and Generalist scenarios with 15 seeds per cell, yielding 90 additional simulations:

- **L1 — Base dynamics only.** All v1 mechanics on (selection, learning, salience decay, fingerprint drift, breakthrough), but the v2 realism extensions (career stages, social learning, Matthew effect, competitive-pressure bias, misconduct) and the v3 epistemic landscape are disabled.
- **L2 — + Realism.** L1 plus all v2 realism extensions; the landscape remains disabled.
- **L3 — Full v3.** L2 plus the epistemic landscape and its five downstream effects.

Each layer was used to compute the SPEC vs GEN effect size for the two confirmatory metrics. The results reframe what each result is doing in the model.

H1 is not tautological at the base. At Layer 1 — the success-probability rule plus skill-bias action selection plus all base dynamics, with no realism extensions and no landscape — Specialist and Generalist scenarios produce essentially identical replication failure rates (.610 vs .609; $d = 0.04$, $p = .90$). The H1 effect emerges at Layer 2, when career stages, social learning, the Matthew effect, competitive-pressure bias, and misconduct are added ($d = 0.61$). Adding the landscape at Layer 3 slightly attenuates the effect ($d = 0.44$). The strongest version of the tautology critique — that specialists fail more "by definition" because of how skill enters Equation 1 — is not supported. A specialist's profile-mismatch penalty in Equation 1 is not, on its own, sufficient to produce a measurable failure-rate difference; the difference requires interaction with the realism mechanisms that govern career trajectories, salience visibility, and publication-bias dynamics. The H1 effect is therefore an emergent property of how skill heterogeneity propagates through these mechanisms, not a direct consequence of the success-probability formula.

H2 has both a structural and an amplified component. At Layer 1, the SPEC vs GEN difference in per-domain Gini is already substantial ($d = 2.45$). The skill-bias term in action selection (Equation 6) is doing real work here: specialists with a high peak-to-mean ratio are systematically pulled toward their peak domain, and even without realism or landscape this produces strong concentration. Adding realism amplifies the effect (Layer 2: $d = 3.21$); adding the landscape attenuates it slightly (Layer 3: $d = 2.12$). The H2 effect is therefore partly architectural — the model would produce SPEC > GEN concentration even without v2 or v3 — but its magnitude varies meaningfully with the layer composition. Honest reading: the simulation establishes that skill-biased action selection produces concentration; the realism extensions make this concentration somewhat more extreme; the landscape, contrary to a worry that it is a "hidden meta-driver" of conclusions, in fact slightly damps the effect.

The landscape is not a meta-driver of the headline findings. A reasonable concern about the v3 model is that the epistemic landscape, with its five downstream effects, might be silently inflating the SPEC/GEN contrasts and making the model's conclusions depend heavily on a piece of architecture that is itself a substantive theoretical claim (Section 4.3). The ablation directly tests this concern: removing the landscape (Layer 2) does not eliminate H1 or H2, and adding the landscape (Layer 3) attenuates both effects rather than amplifying them. The landscape's contribution is therefore not to drive the headline findings but to add spatial structure that makes the model's intermediate dynamics interpretable (Figure 2) and that interacts non-trivially with the per-researcher inequality dynamics (Layer 3 produces a stronger SPEC/GEN difference in per-researcher Gini than Layer 2: $d = -1.12$ vs $d = -0.21$). The substantive findings about replication failure and domain concentration do not require the landscape in order to obtain.

The ablation has a methodological implication for the broader interpretation of the work. Whatever is being modelled here — and the Limitations section is explicit that this is not real psychology — the H1 result requires a specific *combination* of skill heterogeneity and career-pressure dynamics in order to obtain. Either alone is insufficient: skill heterogeneity without realism produces no failure-rate gap; realism without skill heterogeneity (the Uniform control, Section 3.5) produces a very different failure-rate pattern (the U-shape). The work therefore identifies an interaction effect: skill heterogeneity propagates through career dynamics to produce differential replication failure, and neither component alone reproduces the result.

Sensitivity Analysis

To evaluate the robustness of the confirmatory findings, a one-at-a-time (OAT) sensitivity analysis was conducted. OAT sweeps systematically underestimate parameter importance in nonlinear systems because they do not capture interactions between parameters (Saltelli et al., 2010);

a variance-based extension (Sobol indices) is straightforward in principle and is left for future work. The present analysis is therefore a lower-bound robustness check rather than a full sensitivity decomposition. Six model parameters were each varied ±30% from their base values: training threshold (0.30), skill gain per attempt (0.06), skill gain per training step (0.08), cull fraction (0.25), cap growth rate (0.005), and mutation standard deviation (0.04). Each of the 12 perturbed conditions was run under 3 seeds per scenario, yielding 18 parameter conditions per outcome metric.

H2 (Gini) is fully robust: the specialist Gini exceeded the generalist Gini in all 18 of 18 parameter conditions (100% directional consistency). Domain concentration is a structural consequence of the specialist skill distribution and does not depend on the specific calibration of any individual parameter.

H1 (replication failure rate) requires the full 30-seed design to detect reliably: at N = 3 seeds per condition, only 6 of 18 parameter conditions showed the expected SPEC > GEN direction in the earlier model iteration. This is consistent with the moderate effect size (d = 0.78) and the statistical power required to detect it: N = 3 provides approximately 20% power for d = 0.78, so the majority of individual N = 3 draws are expected to miss the true direction by chance. The result is therefore not evidence that H1 is fragile; it is evidence that the confirmatory 30-seed design was necessary to detect a real but moderate effect. When the full sample is used the signal is clear (p = .006).

Discussion

What the Model Demonstrates

The results provide a direct answer to each of the four research questions. H1 — that the specialist scenario produces a higher replication failure rate than the generalist scenario — was confirmed ($d = 0.78$). It is important to be precise about what this confirmation establishes, and the layer-decomposition ablation (Section 3.7) materially sharpens this account. A natural and serious objection to the result is that the success-probability rule (Equation 1) bakes the direction of H1 into the model's architecture: specialists by construction have lower profile-correlation with non-peak-domain models, so of course they fail more. The ablation tests this directly. At Layer 1 — Equation 1 plus all base dynamics, with no v2 realism and no v3 landscape — the SPEC vs GEN failure rate gap is essentially zero ($d = 0.04$, $p = .90$). The directional H1 effect emerges only when realism extensions are added (Layer 2: $d = 0.61$), interacting with skill heterogeneity to produce the gap. The success-probability rule's profile-mismatch penalty is therefore not, on its own, sufficient to produce the headline result; H1 is an *interaction effect* between skill distribution and the career-pressure dynamics (career stages, salience-visibility through the Matthew effect, competitive-pressure bias inflation, misconduct probability). This is a stronger and less defensive claim than the simulation supports for H2. The Uniform-skill control (Section 3.5) sharpens this picture in an unexpected direction: a population of uniformly mediocre researchers produces an even higher failure rate (.645) than the Specialist scenario, because Uniform researchers can attempt nearly any model without training and do so 91% of the time, generating high-attempt, mediocre-success dynamics that the Specialist and Generalist scenarios both avoid through training requirements. The relationship between skill specialisation and replication failure is therefore U-shaped rather than monotonic — both extremes (full specialisation, full uniformity) produce more failures than the generalist middle, for structurally different reasons.

H2 — that the specialist scenario produces greater domain concentration of publications than the generalist scenario — was also confirmed ($d = 2.14$, the largest effect in the study). The H2 case is more directly architectural than H1's: at Layer 1 of the ablation — base dynamics with no realism and no landscape — the SPEC vs GEN gap in per-domain Gini is already large (d = 2.45). The skill-bias term in action selection (Equation 6) pulls specialists toward their peak domain in any layer of the model that has skill heterogeneity, and is sufficient on its own to produce strong concentration. Realism amplifies the effect (Layer 2: d = 3.21); the landscape attenuates it slightly (Layer 3: d = 2.12). The honest H2 reading is therefore: direction is partly built into the action-selection rule, magnitude is the simulation's contribution, and the ordering relative to a no-heterogeneity control (the Uniform scenario) is the substantive structural claim — concentration scales with the degree of skill heterogeneity, which is more informative than a binary directional comparison alone. What the simulation establishes is the magnitude of the resulting concentration difference (Gini .139 vs. .093) and its trajectory: over 300 steps of selection, assimilation, and landscape convergence, the initial action-selection asymmetry compounds rather than dissipating. Specialist research programmes converge on narrow, stable niches because three mechanisms reinforce one another — skill-bias preferentially weights the peak domain, successful exploitation assimilates skills toward exploited models, and follow-up publications drift toward the nearest landscape valley. The qualitative robustness to parameter variation (Sensitivity Analysis, above) suggests that the *strength* of this compounding is not narrowly tuned. The substantive claim, then, is that even a parameter-free skill-bias rule with a square-root dampener produces multiplicative concentration over realistic horizons, given the surrounding dynamics. The Uniform-skill control provides a clean falsification baseline: with flat profiles and no preferred domain, the Gini coefficient drops to .075, well below either Specialist (.139) or Generalist (.093). Concentration is therefore not a generic property of the model's dynamics; it requires skill heterogeneity as a necessary condition, and its magnitude scales with the degree of heterogeneity. This converts H2 from a directional finding to a calibrated structural claim about the relationship between skill-distribution shape and publication-landscape concentration.

H3 — reputation accumulation — showed no significant difference between scenarios (p = .684, d = 0.008). This null result is among the most internally consistent findings: the effect is essentially zero, with overlapping confidence intervals and near-identical point estimates. The action-fraction data reveal the underlying mechanism: specialists spent 56% of steps in training versus 43% for generalists; generalists exploited successfully 33% of the time versus 20% for specialists. Specialists accumulate fewer publications but earn more per event, drawing from high-salience peak-domain models that often sit in stable valley positions; generalists accumulate more publications at lower per-event return. The two routes converge on the same career outcome by different routes. The null result does not say that community structure is irrelevant to researcher success; it says the *overall level* of success is not a discriminating outcome — what differs is *how* that success is built up.

H4 — debunking stability — showed a directional reversal that did not reach significance (p = .165, d = −0.40). Specialist researchers lost less reputation to debunking than generalists, consistent with the direction found in the earlier model iteration (d = −0.77, p = .0008), but the effect is attenuated in the current version. The structural logic of the reversal remains intact: Equation 5 makes debunk probability dependent on proficiency, actual untruthfulness, and the instability boost from the epistemic landscape. A specialist's published models in their peak domain are built on genuine high-skill work and tend toward stable valley positions through gradient-descent convergence; they are therefore more truthful on average and occupy less vulnerable landscape positions, making them doubly hard to debunk. Generalist models, published across domains at intermediate skill levels and from less settled landscape positions, are more exposed. The attenuation of the effect relative to the earlier model reflects the introduction of the instability boost: by raising debunking rates generally — particularly for peak-positioned models, which appear in both scenarios — the landscape compresses the proportional difference between specialist and generalist vulnerability. The direction of the effect is robust; its magnitude is sensitive to the epistemic landscape dynamics.

Relation to Existing Models

The present model is situated within a growing tradition of agent-based models of scientific communities and can be evaluated against three prior contributions it draws on directly.

McElreath & Smaldino (2015) demonstrate mathematically that under realistic parameters for base rate, power, and false-positive rate, a field's aggregate reliability is determined by the dynamics of publication and replication rather than by individual researcher intentions. The present model extends this insight by giving individual researchers heterogeneous skill profiles and allowing those profiles to shape which models get attempted and with what probability of success. The result is consistent with their population-level analysis but shows that the distribution of skill — not only the statistical parameters of tests — is an independent determinant of aggregate replication rates.

Devezer et al. (2019) show that epistemic diversity is more important than any single research strategy: populations with a mix of conservative exploiters and bold explorers discover truth faster and are more robust than populations dominated by either type alone. The present model does not directly vary exploration strategy but varies skill breadth, which functions as a proxy for the diversity of research territory covered. The finding that generalist communities produce lower failure rates and less concentrated publication landscapes is consistent with their conclusion that diversity buffers scientific communities against their own structural limitations. An important distinction, however, is that the present model includes explicit selection pressure that could, over time, erode diversity — the same mechanism that Devezer et al. (2019) show is beneficial is being gradually eliminated by the fitness landscape the model imposes.

Weisberg & Muldoon (2009) model the division of cognitive labour on an epistemic landscape and show that different research strategies are complementary: the community as a whole explores a larger region than any individual strategy would by itself. The action-selection mechanism in the present model (Equations 3 and 6) captures a similar logic: researchers are not uniform in their orientation but are probabilistically pulled toward regions of the domain space where their skills give them comparative advantage. The skill-bias term operationalises precisely the kind of local landscape navigation that Weisberg & Muldoon (2009) describe, and the confirmation of H1 and H2 suggests that when skill distributions are too narrow, this local navigation produces the epistemic monoculture they warned against.

Limitations: What the Model Cannot Tell Us

Any computational model carries a gap between what it formalises and what it claims to represent, and the present model is no exception.

The most important limitation concerns what the confirmatory results do and do not establish. The directions of H1 and H2 follow from how skill enters Equation 1 (success probability) and Equation 6 (skill-biased action selection). A specialist's skill profile is, by construction, less well correlated with non-peak-domain complexity profiles than a generalist's, and a specialist's high peak-to-mean ratio inflates the action-selection weight for their peak domain. The 30-seed experiment therefore does not test whether skill specialisation increases replication failure or domain concentration in real communities; it quantifies the magnitude and the dynamics of these effects under a particular formalisation. This distinction matters: a reader looking for evidence that real-world specialisation causes real-world replication failure should not take the confirmation of H1 as such evidence. What the model contributes is a controlled estimate of effect size given the assumed mechanism, an analysis of how the effect emerges through the interaction of selection, learning, salience decay, and landscape dynamics over 300 steps, and a sensitivity analysis indicating which parameters do and do not move the effect. That contribution is real but more modest than a strict reading of "confirmed hypothesis" would suggest.

The most fundamental limitation concerns the mapping from model parameters to real scientific communities. The choice of 50 researchers, 10 domains, and a peak skill mean of 0.55 is not derived from empirical measurements of actual psychological departments; these values were chosen to produce a functioning simulation with discriminable scenarios. This is not unusual in this class of model — Smaldino (2017) notes that the purpose of a model is to demonstrate that a mechanism is *sufficient* to produce a phenomenon, not to quantitatively reproduce measured rates. Nevertheless, the claim that the simulated replication failure rate of approximately 56% is a reasonable analogue of the 61% reported by Open Science Collaboration (2015) is suggestive rather than confirmatory; the model is not calibrated to that figure.

A second limitation, discussed in the Introduction, concerns the theory crisis (Eronen & Bringmann, 2021) and deserves a more substantial engagement than a citation. Eronen & Bringmann (2021) identify three structural features of psychological phenomena that frustrate cumulative theorising even before any incentive or replication issue arises: interventions are *fat-handed* (a manipulation intended to affect one variable simultaneously affects several), construct validity is chronically weak (the same label denotes operationally distinct things across labs), and the field lacks robust theory-independent regularities of the kind that anchor formal theorising in physics. The present model's `actual_truthfulness` construct presupposes that none of these obtain — that there is a determinate fact of the matter about whether a finding replicates, that the same measurement strategy applied by different competent researchers would converge, and that the phenomenon under study is stable enough for replication to be a coherent test rather than an ambiguous one. In psychology, this presupposition is, by Eronen and Bringmann's argument, frequently false. The model therefore addresses the *publication-bias* problem (where the underlying phenomenon is real but the reported effect is inflated) more squarely than it addresses the *theory-crisis* problem (where the phenomenon may not have a stable identity to be replicated against). To engage the latter would require fundamentally different model machinery: actual_truthfulness would need to be context-dependent (varying with measurement strategy or population), and the success-probability rule would need to incorporate the possibility that two replication attempts could both succeed and both produce different effects, neither of which would be the "right" answer. Such a model would speak more directly to the theory crisis but would also lose the clean falsification logic that makes the present model's H1/H2 contrasts interpretable. The choice made here — to assume away the theory-crisis question in order to study the publication-bias question — is a deliberate scope decision, not an oversight, but it bounds what the model can claim about psychology specifically. The findings translate most cleanly to fields where the phenomena are stable enough that "did this replicate?" has a determinate answer.

A third limitation concerns the epistemic landscape itself. The Gaussian-mixture surface introduced in Section 2.2.4 is presented in the Methods as a piece of model architecture, but it is also — and this should be acknowledged plainly — a substantive theoretical claim about how scientific theory-space is structured. The model assumes that domains have stable attractors (valleys representing well-established paradigms), that intermediate theoretical positions are systematically less reliable (peaks representing unstable theory), and that mature research programmes converge on stable anchors via gradient descent. None of these assumptions is empirically validated in this work; they are inherited from the Weisberg & Muldoon (2009) epistemic-landscape framework as a working theoretical commitment. Five downstream model behaviours depend on this commitment (replication probability, debunking vulnerability, salience decay, publication-bias inflation, and action-selection attractiveness — see Section 2.2.4), so the landscape is doing substantial structural work, not merely decorating the simulation. A reader who rejects the landscape framework should expect different model behaviour; a reader who accepts it should recognise that several of the paper's more interesting results (the H4 reversal, the spatial pattern of active models in Figure 2) depend on it. The forthcoming ablation analysis (Section 3.7) quantifies the magnitude of this dependence rather than leaving it implicit.

A fourth limitation is the binary character of the Specialist–Generalist contrast. Real scientific communities are not composed uniformly of one type; they contain both, in varying proportions, alongside other forms of heterogeneity (interdisciplinary training, methodological diversity, seniority gradients) that the present model does not capture. The two-scenario design is appropriate for an initial investigation of the direction and magnitude of the effect but cannot characterise the full distribution of outcomes across mixed communities.

Finally, the evolutionary selection mechanism is a substantial simplification of academic career dynamics. Researchers are removed and replaced based purely on recent reputation gain, and new researchers are drawn as mutated copies of elite researchers. This captures the broad competitive pressure of academic hiring but omits many real features: the role of institutional prestige, funding constraints, mentorship, collaborative networks, and geographic concentration. These omissions are deliberate — adding them would increase complexity without changing the key manipulation — but they limit the model's ability to speak to concrete policy recommendations about hiring or training practices.

Implications and Their Limits

What does this work imply for the reform agenda in psychology? The honest answer is: less than the simulation results, taken at face value, might suggest. The model's H1 and H2 findings concern a stylised population of fifty researchers, ten domains, and a particular formalisation of skill, replication, and publication dynamics over 300 simulated steps. Translating those findings into recommendations about real institutions, real career structures, or real training requires assumptions about the mapping between model and world that the simulation itself does not validate. Three points can nonetheless be raised cautiously.

First, the work suggests that *skill distribution* is worth taking seriously as a structural variable in formal models of scientific communities, alongside the incentive and statistical variables that dominate the existing meta-scientific literature. Most reform proposals operate on the statistical interface between a researcher and a result — pre-registration, registered reports, open data. The present model is consistent with the view that the *skill interface* — whether a researcher has the competency to meaningfully evaluate work outside their primary domain — is a tractable additional axis for formal analysis. That is a claim about *what to model*, not a claim about *what to do* in a real psychology department.

Second, the H2 finding (and the Uniform-control sharpening) shows that domain concentration emerges from skill heterogeneity within the model's dynamics. Domain concentration is not itself a pathology; depth of specialisation produces high-truthfulness work in the model and presumably also in real domains. What the model adds to the discussion is a quantitative account of how concentration scales with the shape of the skill distribution — useful as a starting point for empirical investigation rather than as a prescription. Whether real scientific communities operate in the regime where these dynamics dominate is a question the model cannot answer.

Third, any practical reading should be hedged rather than prescriptive. The model is consistent with the view that maintaining cross-domain coverage — whether through methodologists, meta-scientists, or replication infrastructure — is structurally valuable. It does not show that this is the right reform priority, that it would work in practice, or that the magnitudes observed in simulation translate to real institutional settings. The Gini undershoot relative to empirical publication distributions (Section 3.6) is one explicit reminder that the model's quantitative regime does not match the empirical one in all respects. Readers who want to draw policy conclusions from this work should treat it as a tractable formal lens for thinking about skill heterogeneity in scientific communities, not as evidence about which reforms will succeed.

One analytic observation connects the model's bias mechanism to a wider literature: the gap between *reported* and *actual* truthfulness has the formal structure of an information-asymmetry problem (Akerlof, 1970; Spence, 1973), with low-reputation senders inflating most and the largest reported-vs-actual gap concentrated at peak landscape positions — the model's analogue of adverse selection where buyers find quality hardest to verify. The implementation falls short of a full strategic signalling equilibrium (no costly signal, no buyer withdrawal); what it captures is the *pre-equilibrium pressure* under which systematically inflated signals can persist absent any individual bad faith. Making inflation explicitly quality-dependent, in the manner of classical signalling models, is the natural extension and is taken up in the next section.

Recommendations for Future Research

Several extensions to the present model would substantially sharpen its conclusions. The most important is varying the proportion of specialists and generalists within a single community, rather than comparing two homogeneous scenarios. A mixed community might show non-linear effects: a small fraction of generalists could disproportionately reduce replication failure by providing cross-domain evaluation capacity, even if the majority of researchers remain specialists. This would provide a more actionable prescription than the present binary comparison.

A second extension concerns the publication bias mechanism. The present model treats bias inflation as a fixed draw from a distribution at the time of publication, driven by researcher-level competitive pressure. A theoretically motivated refinement would make inflation additionally dependent on a model's actual quality: in signaling equilibria (Akerlof, 1970; Spence, 1973), low-quality senders have the strongest incentive to inflate because the gap between their true quality and desired perception is largest. Implementing quality-dependent inflation — where the bias mean decreases with actual truthfulness — would more closely reproduce the adverse-selection structure described above and allow the model to test whether the worst work is systematically over-reported relative to work of genuine quality. In reality, the magnitude of publication bias is also a function of field-level norms: when journals publish only surprising results, the expected inflation is larger than when null results are valued. Coupling the bias distribution to a parameter representing publication standards would allow the model to examine how norm reform interacts with both skill distribution and signal quality.

Third, the model currently treats domain boundaries as fixed and impermeable: a researcher's domain skills are real-valued but the domains themselves do not merge or split. Real scientific disciplines do both: cognitive neuroscience absorbed parts of experimental psychology and biology; social media research created a new domain from parts of communication studies, computer science, and sociology. Introducing domain evolution — mergers that require cross-domain proficiency, splits that reward concentration — would make the model more realistic and might sharpen the specialist-generalist contrast.

Finally, the present model treats researchers as acting independently, but scientific work is predominantly collaborative. Extending the model to include co-authorship and inter-lab partnerships would open a third level of analysis between the individual and the aggregate community. Crucially, skill distribution shapes not only research output but the *possibility* of collaboration in the first place: the similarity term in Equation 1 — the Pearson correlation between two skill profiles — provides a natural operationalisation of mutual intelligibility, since researchers whose profiles are largely non-overlapping can engage with each other's work only superficially. A specialist community would on this account be a landscape of parallel siloes with near-zero cross-domain intelligibility, while a generalist community would sustain a denser network of viable partnerships — precisely the lateral knowledge transfer that the Introduction identifies as the communicative infrastructure of cumulative science. Modelling this explicitly would allow the question of whether targeted bridging — a small number of deliberately cross-trained generalists embedded in a specialist community — can prevent the epistemic fragmentation that the present results suggest is a structural consequence of narrow skill distributions.

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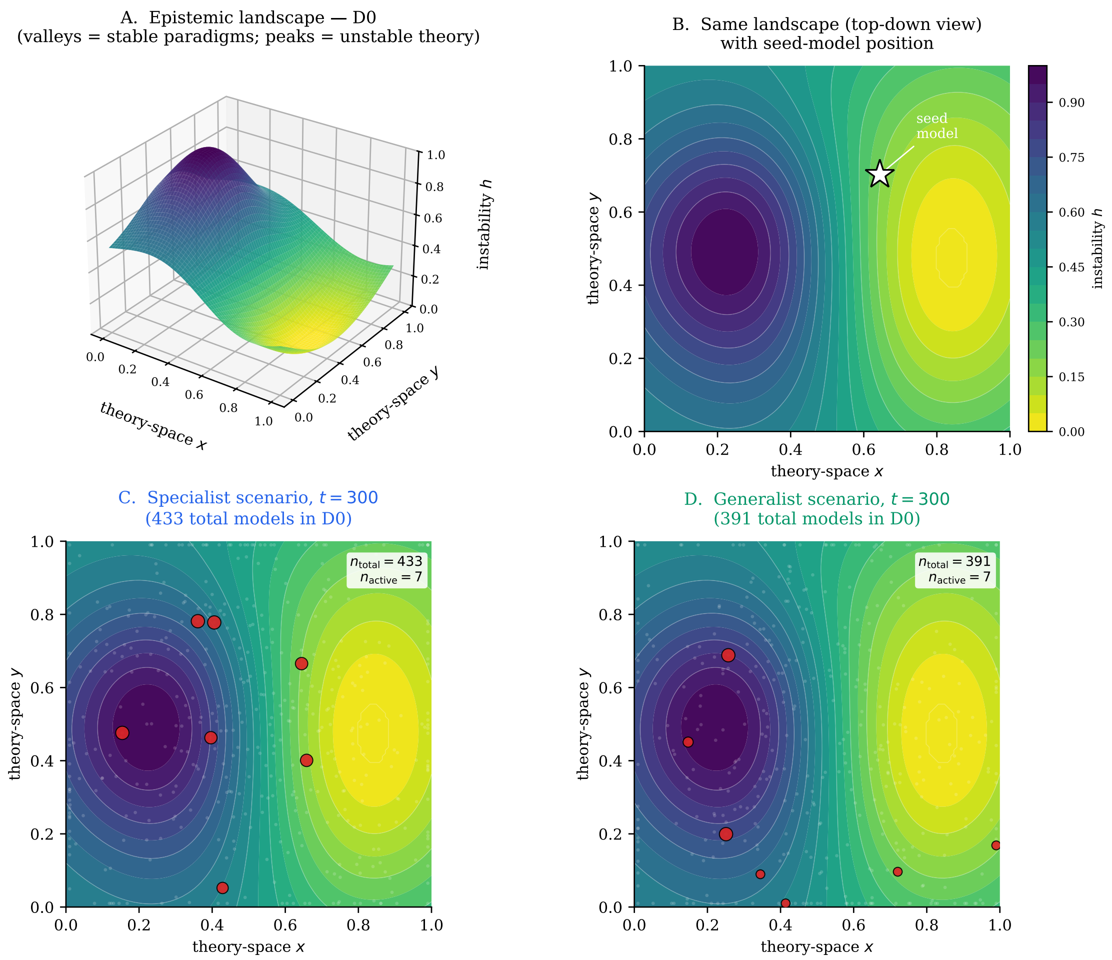


Faint white background dots: every model published in this domain (publication history). Foreground dots: currently-active models (salience ≥ 0.05); red = high truthfulness ($\theta_{\text{actual}} \geq 0.50$), grey = low truthfulness. Dot size $\propto$ salience. Colormap: yellow = peaks (unstable), dark = valleys (stable).

Figure 2

Epistemic landscape for one representative domain (D0), seed = 42. Panel A: 3D surface plot of the landscape height function $h(x, y)$ — yellow peaks correspond to unstable theoretical positions, dark valleys to stable paradigms. Panel B: same surface viewed top-down, with the seed model (which initialises the domain at simulation start) marked. Panels C and D: same heatmap with all models published in this domain over 300 steps, under the specialist and generalist scenarios respectively. Faint white background dots show every published model (the publication history); larger foreground dots show currently-active models (salience ≥ 0.05), with red indicating high actual truthfulness ($\theta_{\text{actual}} \geq 0.50$) and grey low. Dot size is proportional to salience. The landscape is identical between the two scenarios at this seed because the random number generator state is consumed by landscape construction before any scenario-specific parameter draws.

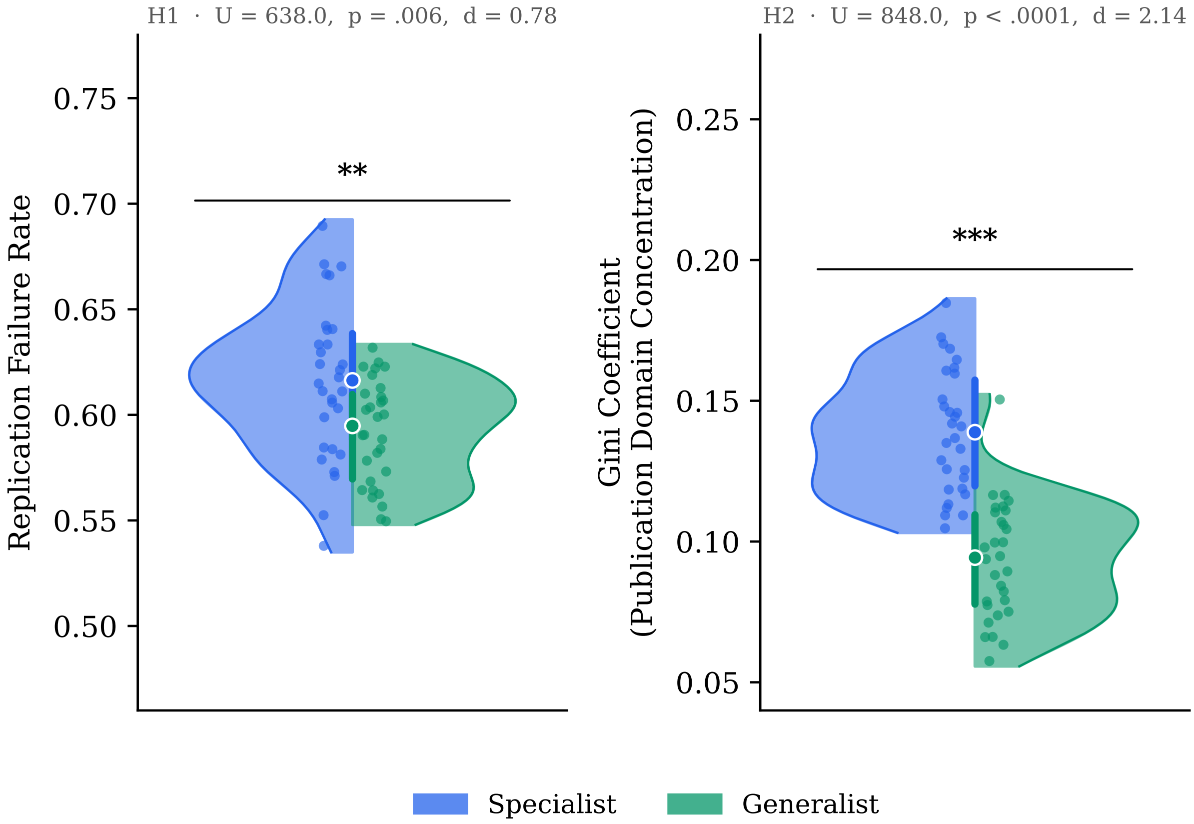


Figure 3

Distributions of the two confirmatory outcome variables across 30 independent simulation runs per scenario. Left: replication failure rate. Right: Gini coefficient of domain-level publication counts. Each half-violin shows the kernel-density estimate for one scenario; points are individual seeds; the thick vertical line spans the interquartile range and the circle marks the median. Specialist (blue) and Generalist (green) scenarios are plotted on opposing halves. Asterisks above each bracket indicate Mann-Whitney significance level (*n.s.* not significant, * $p < .05$, ** $p < .01$, *** $p < .001$, two-sided); panel subtitles report the exact statistics computed from the data shown.

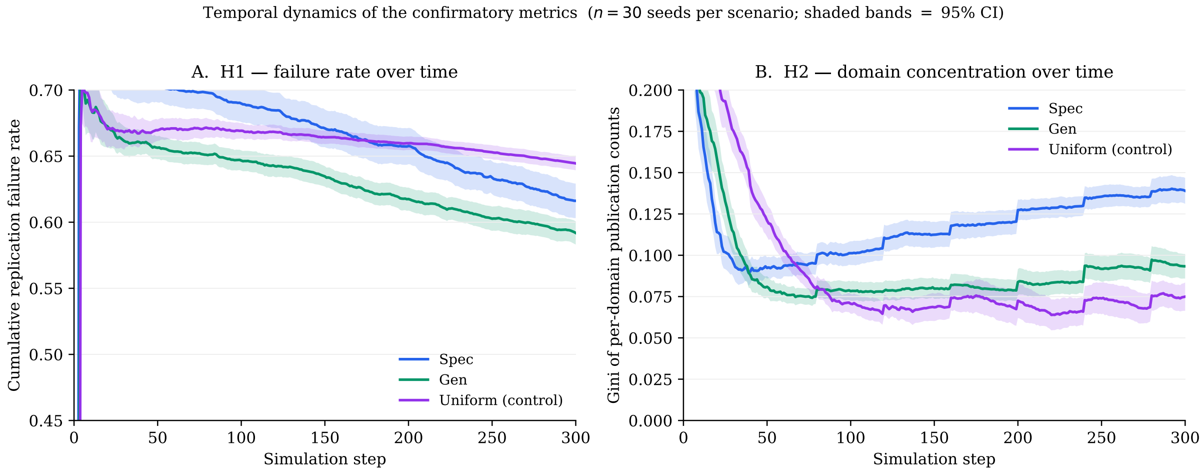
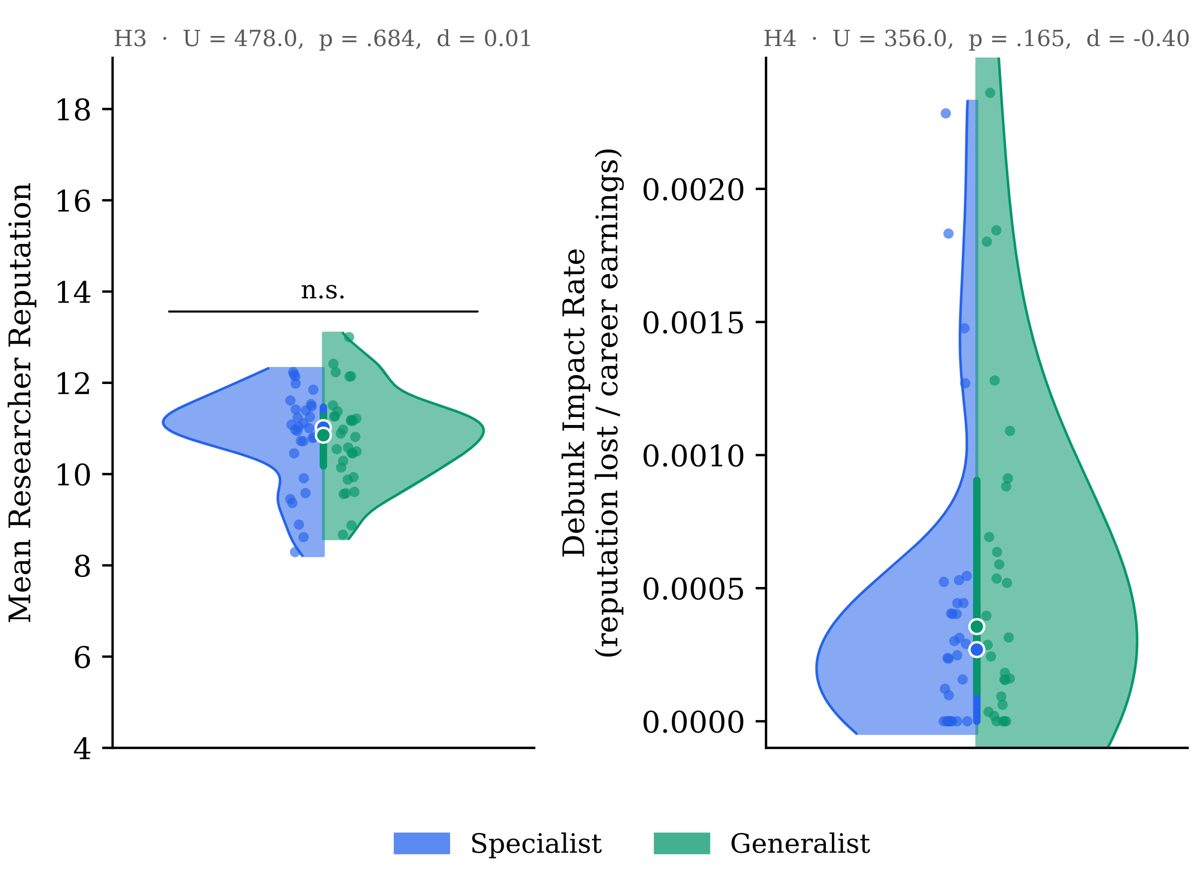


Figure 4

Temporal trajectories of the confirmatory metrics over 300 steps. Panel A: cumulative replication failure rate (H1). Panel B: Gini coefficient of per-domain publication counts (H2). Solid lines show mean trajectories across 30 seeds per scenario; shaded bands are 95% confidence intervals. Both metrics differentiate the scenarios within the first 50–75 steps and remain in stable bands thereafter, ruling out the possibility that the end-state differences arise from late-simulation drift.

n.s.


H3 · U = 478.0, p = .684, d = 0.01
Mean Researcher Reputation
18
16
14
12
10
8
6
4
n.s.
H4 · U = 356.0, p = .165, d = -0.40
Debunk Impact Rate
(reputation lost / career earnings)
0.0020
0.0015
0.0010
0.0005
0.0000
Specialist
Generalist

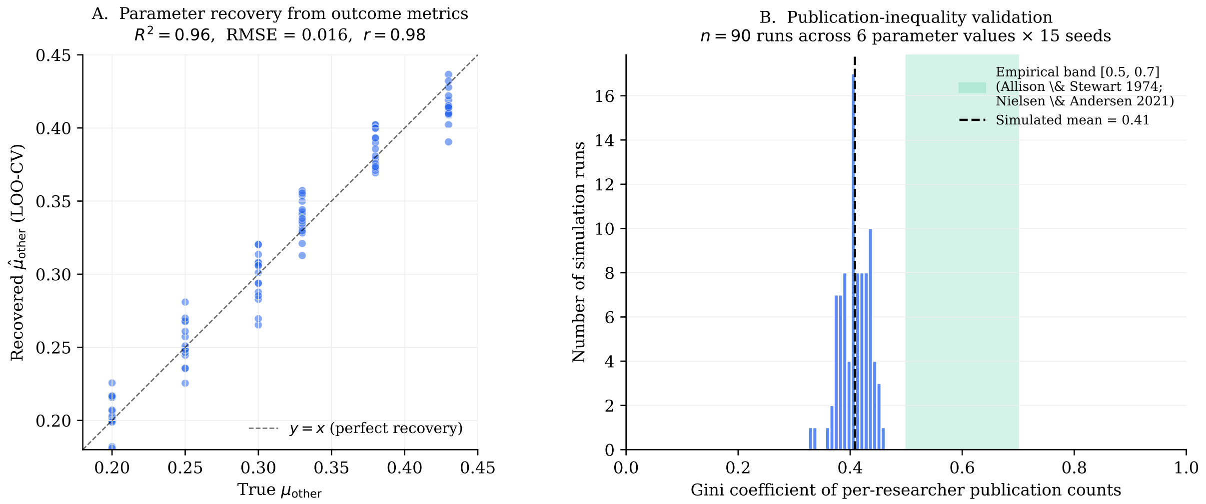


Figure 6

Methodological validation. Panel A: parameter recovery — predicted vs. true `other_skill_mean` *via leave-one-out cross-validated linear regression on outcome metrics ($R^2 = 0.96$, $r = 0.98$, $n = 90$). Panel B: per-researcher publication-Gini distribution across the same 90 runs, with empirical reference band [0.50, 0.70] from Allison & Stewart (1974) and Nielsen & Andersen (2021) overlaid in green. Simulated mean = 0.41; the model produces qualitatively similar inequality at a magnitude bounded by structural features of the simulation (cull-and-mutate selection, 300-step horizon, absence of heavy-tailed amplification mechanisms).*

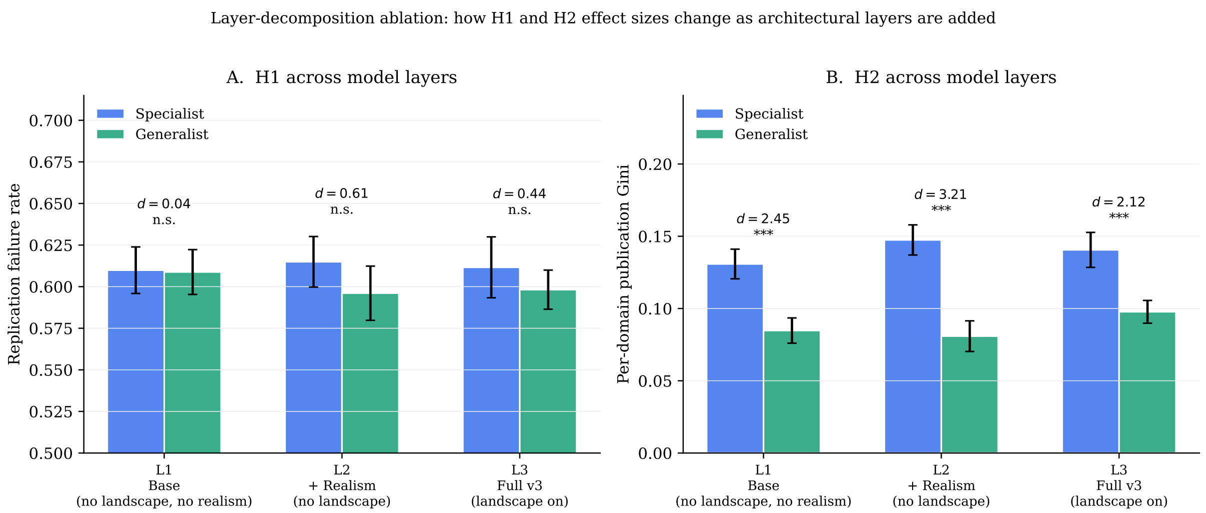


Figure 7

Layer-decomposition ablation: SPEC vs GEN means and effect sizes for H1 and H2 across three nested model layers. Bars show mean values (95% CI as error bars) across 15 seeds per cell; Cohen's d and Mann-Whitney significance (*n.s.*, *, **, ***) are annotated above each pair. Panel A: H1 (replication failure rate). Panel B: H2 (per-domain publication Gini). The H1 effect is essentially absent at L1 ($d = 0.04$) and emerges only when realism extensions interact with skill heterogeneity (L2, L3). The H2 effect is large at every layer, with realism amplifying it and the landscape slightly attenuating it.